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Systems and methods for providing power to pulse-width-modulation controllers of power converters during normal operation

Abstract

Controller and method for a power converter. For example, the controller includes: a first terminal configured to receive a first voltage; a second terminal connected to a capacitor and biased to a second voltage; a voltage detector configured to receive the second voltage from the second terminal and generate a detection signal based at least in part on the second voltage; a charging controller configured to receive the detection signal and generate a first control signal based at least in part on the detection signal; and a charging current generator configured to receive the first voltage from the first terminal and receive the first control signal from the charging controller; wherein the voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the detection signal indicating that the second voltage has decreased to the first predetermined threshold.

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Background/Summary

1. CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/131,788, filed Apr. 6, 2023, which is a continuation of U.S. patent application Ser. No. 17/465,678, filed Sep. 2, 2021, which claims priority to Chinese Patent Application No. 202010919275.0, filed Sep. 4, 2020, all of the above applications being incorporated by reference herein for all purposes.

2. BACKGROUND OF THE INVENTION

(1) Certain embodiments of the present invention are directed to circuits. More particularly, some embodiments of the invention provide systems and methods for providing power to pulse-width-modulation controllers of power converters. Merely by way of example, some embodiments of the invention have been applied to power converters during normal operation. But it would be recognized that the invention has a much broader range of applicability.

(2) A conventional power converter often includes a semiconductor chip. For the semiconductor chip, a high-voltage pin (e.g., an HV pin) is generally connected to one or more AC terminals of the power converter through one or more components such as one or more diodes and/or is connected to one or more capacitors (e.g., a bulk capacitor) of the power converter. During the startup process of the semiconductor chip, the high-voltage pin (e.g., an HV pin) charges an external capacitor (e.g., a VDD capacitor) through an internal charging circuit of the chip.

(3) After the voltage (e.g., a VDD voltage) provided by the external capacitor rises to a chip startup threshold, the internal charging circuit is disconnected and the startup process of the semiconductor chip ends. Upon completion of the startup process, the semiconductor chip starts normal operation for the power converter. As the output voltage of the power converter rises, an auxiliary winding of the power converter starts providing power to the semiconductor chip. Often, such start-up process that uses the high-voltage pin (e.g., an HV pin) can speed up the startup and also reduce power consumption of the semiconductor chip after the startup.

(4) FIG. 1 is a simplified diagram showing a conventional power converter. The power converter

100 (e.g., an AC-DC power supply) includes a primary winding **102**, a secondary winding **104**, an auxiliary winding **106**, a transistor **110** (e.g., a transistor M1), a rectifier **120** (e.g., a bridge rectifier), a filter **122** for electromagnetic interference (EMI), diodes **124**, **126**, **162**, **170** and **176**, an optocoupler **128**, a pulse-width-modulation (PWM) controller **130** (e.g., an IC chip), resistors **152**, **154**, **160**, **164**, **166**, **172**, **174**, **184**, **186**, **188** and **196**, capacitors **156**, **158**, **168**, **178** and **194**, and an error amplifier **198** (e.g., TL431). The pulse-width-modulation (PWM) controller **130** (e.g., an IC chip) includes a terminal **132** (e.g., a PRT pin), a terminal **134** (e.g., an FB pin), a terminal **136** (e.g., a CS pin), a terminal **138** (e.g., a GND pin), a terminal **140** (e.g., a HV pin), a terminal **142** (e.g., an NC pin), a terminal **144** (e.g., a VDD pin), and a terminal **146** (e.g., a GATE pin). Additionally, the optocoupler **128** includes a light emitting diode **192** and a photosensitive detector **190**. Moreover, the terminal **134** (e.g., an FB pin) is used to receive a feedback signal **135** that represents an output voltage **181** (e.g., V.sub.out) of the power converter **100**, and the terminal **140** (e.g., a HV pin) is used to receive a voltage **141**.

(5) As shown in FIG. 1, the EMI filter **122** receives an alternating current (AC) voltage **150**. One output terminal of the EMI filter **122** is connected to an anode of the diode **124** (e.g., a diode D1), and another output terminal of the EMI filter **122** is connected to an anode of the diode **126** (e.g., a diode D2). A cathode of the diode **124** (e.g., a diode D1) and a cathode of the diode **126** (e.g., a diode D2) both are connected to one terminal of the resistor **152** (e.g., a resistor R2), which includes another terminal that is connected to one terminal of the resistor **154** (e.g., a resistor R3). Another terminal of the resistor **154** (e.g., a resistor R3) is connected to the terminal **140** (e.g., a HV pin) of the PWM controller **130**.

(6) The anode of the diode **124** (e.g., a diode D1) is connected to an input terminal of the rectifier **120** (e.g., a bridge rectifier), and the anode of the diode **126** (e.g., a diode D2) is connected to another input terminal of the rectifier **120** (e.g., a bridge rectifier). The rectifier **120** (e.g., a bridge rectifier) includes four diodes (e.g., diodes D3, D4, D5, and D6). One output terminal of the rectifier **120** (e.g., a bridge rectifier) is connected to one plate of the capacitor **156** (e.g., a capacitor C2) that is biased to a ground voltage on the primary side, and another output terminal of the rectifier **120** (e.g., a bridge rectifier) is connected to another plate of the capacitor **156** (e.g., a capacitor C2) that is connected to one terminal of the resistor **160** (e.g. a resistor R1), one terminal of the capacitor **158** (e.g., a capacitor C3), and one terminal of the primary winding **102**. Another terminal of the resistor **160** (e.g. a resistor R1) and another terminal of the capacitor **158** (e.g., a capacitor C3) both are connected to a cathode of the diode **162** (e.g., a diode D7). An anode of the diode **162** (e.g., a diode D7) is connected to a drain terminal of the transistor **110** (e.g., a transistor M1) and another terminal of the primary winding **102**.

(7) A gate terminal of the transistor **110** (e.g., a transistor M1) receives a drive signal **147** from the terminal **146** (e.g., a GATE pin) of the PWM controller **130** (e.g., an IC chip), and a source terminal of the transistor **110** (e.g., a transistor M1) is connected to the terminal **136** (e.g., a CS pin) of the PWM controller **130** (e.g., an IC chip) and also connected to one terminal of the resistor **164** (e.g., a resistor R7). Another terminal of the resistor **164** (e.g., a resistor R7) is biased to the ground voltage on the primary side. The terminal **144** (e.g., a VDD pin) of the pulse-width-modulation (PWM) controller **130** (e.g., an IC chip) is connected to one terminal of the resistor **166** (e.g., a resistor R6), and another terminal of the resistor **166** (e.g., a resistor R6) is connected to one plate of the capacitor **168** (e.g., a capacitor C1) and a cathode of the diode **170** (e.g., a diode D8). Another plate of the capacitor **168** (e.g., a capacitor C1) is biased to the ground voltage on the primary side. An anode of the diode **170** (e.g., a diode D8) is connected to one terminal of the auxiliary winding **106**, and another terminal of the auxiliary winding **106** is biased to the ground voltage on the primary side.

(8) The anode of the diode **170** (e.g., a diode D8) is connected to one terminal of the resistor **172** (e.g., a resistor R4), and another terminal of the resistor **172** (e.g., a resistor R4) is connected to the terminal **132** (e.g., a PRT pin) of the PWM controller **130** (e.g., an IC chip) and also connected to

one terminal of the resistor **174** (e.g., a resistor R5). Another terminal of the resistor **174** (e.g., a resistor R5) is biased to the ground voltage on the primary side. One terminal of the secondary winding **104** is connected to an anode of the diode **176** (e.g., a diode D9), and a cathode of the diode **176** (e.g., a diode D9) is connected to an output terminal **180** and one terminal of the capacitor **178** (e.g., a capacitor C4). Another terminal of the secondary winding **104** is connected to an output terminal **182** and another terminal of the capacitor **178** (e.g., a capacitor C4). The output terminal **180** provides the output voltage **181** (e.g., V.sub.out) and an output current **183** (e.g., I.sub.out), and the output terminal **182** is biased to a ground voltage on the secondary side.

(9) The output terminal **180** is connected to one terminal of the resistor **184** (e.g., a resistor R10), and another terminal of the resistor **184** (e.g., a resistor R10) is connected to one terminal of the resistor **196** (e.g., a resistor R9), one terminal of the resistor **186** (e.g., a resistor R11), and an input terminal of the error amplifier **198**. Another terminal of the resistor **196** (e.g., a resistor R9) is connected to one terminal of the capacitor **194** (e.g., a capacitor C5), which includes another terminal connected to a cathode of the light emitting diode **192** (e.g., a light emitting diode D10) and connected to an output terminal of the error amplifier **198**. Another terminal of the resistor **186** (e.g., R11) is biased to the ground voltage on the secondary side, and another input terminal of the error amplifier **198** is biased to a reference voltage **197**. An anode of the light emitting diode **192** (e.g., a light emitting diode D10) is connected to one terminal of the resistor **188** (e.g., a resistor R8), and another terminal of the resistor **188** (e.g., a resistor R8) is connected to the output terminal **180**. The terminal **138** (e.g., a GND pin) of the PWM controller **130** (e.g., an IC chip) is biased to the ground voltage on the primary side.

(10) As shown in FIG. 1, after the EMI filter **122** receives the alternating current (AC) voltage **150**, the voltage **141** (e.g., an AC voltage) is provided to the terminal **140** (e.g., a HV pin) of the PWM controller **130** (e.g., an IC chip) so that the capacitor **168** (e.g., a capacitor C1) is charged through the terminal **144** (e.g., a VDD pin). If a voltage **145** at the terminal **144** (e.g., a VDD pin) rises above a startup threshold, the PWM controller **130** (e.g., an IC chip) disconnects the charging circuit inside the controller **130** and starts normal operation for the power converter **100**. Under the normal operation, the output voltage **181** (e.g., V.sub.out) rises so that the auxiliary winding **106**, together with the diode **170** (e.g., a diode D8), the capacitor **168** (e.g., a capacitor C1) and the resistor **166** (e.g., a resistor R6), provides power supply to the terminal **144** (e.g., a VDD pin). The voltage **145** at the terminal **144** (e.g., a VDD pin) of the PWM controller **130** (e.g., an IC chip) is determined as follows:

$$(11) \quad V_{DD} \approx \frac{N_{aux}}{N_s} \times V_{out} \quad (\text{Equation 1})$$

where V.sub.DD represents the voltage **145** at the terminal **144** (e.g., a VDD pin), and V.sub.out represents the output voltage **181** at the output terminal **180**. Additionally, N.sub.aux represents the number of turns of the auxiliary winding **106**, and N.sub.s represents the number of turns of the secondary winding **104**.

(12) FIG. 2 is a simplified diagram showing certain conventional components of the PWM controller **130** of the power converter **100** as shown in FIG. 1. The PWM controller **130** (e.g., an IC chip) includes a charging controller **210**, a circuit **220** for undervoltage-lockout (UVLO), an internal control circuit **230**, a transistor **240** (e.g., a transistor M1), and a diode **250** (e.g., a diode D1).

(13) The circuit **220** for undervoltage-lockout (UVLO) is used to perform the power-up detection and/or the power-down detection for the PWM controller **130** (e.g., an IC chip). If the circuit **220** for undervoltage-lockout (UVLO) detects that the voltage **145** at the terminal **144** (e.g., a VDD pin) rises above the startup threshold, the circuit **220** for undervoltage-lockout (UVLO) outputs a signal **221** (e.g., AVDD) to the internal control circuit **230** in order to power on other components of the PWM controller **130** (e.g., an IC chip), where the signal **221** (e.g., AVDD) indicates the end of the startup process. If the circuit **220** for undervoltage-lockout (UVLO) detects that the voltage **145** at

the terminal **144** (e.g., a VDD pin) falls below an undervoltage threshold, the circuit **220** for undervoltage-lockout (UVLO) outputs the signal **221** (e.g., AVDD) to the internal control circuit **230** in order to power off other components of the PWM controller **130** (e.g., an IC chip). As an example, the internal control circuit **230** includes various control circuits, and the internal control circuit **230** does not include the charging controller **210**. For example, the internal control circuit **230** includes one or more control circuits for one or more oscillators, one or more control circuits for one or more timers, one or more drivers, one or more logic control circuits, and/or one or more protection circuits.

(14) A drain terminal of the transistor **240** (e.g., a transistor M1) is connected to the terminal **140** (e.g., a HV pin) of the PWM controller **130** (e.g., an IC chip), a gate terminal of the transistor **240** (e.g., a transistor M1) is connected to an output terminal of the charging controller **210**, and a source terminal of the transistor **240** (e.g., a transistor M1) is connected to an anode of the diode **250**. A cathode of the diode **250** is connected to the terminal **144** (e.g., a VDD pin) of the PWM controller **130** (e.g., an IC chip) and also connected to an input terminal of the circuit **220** for undervoltage-lockout (UVLO). At an output terminal of the circuit **220** for undervoltage-lockout (UVLO), the signal **221** (e.g., AVDD) is sent to an input terminal of the internal control circuit **230**.

(15) As shown in FIG. 1 and FIG. 2, after the EMI filter **122** receives the AC voltage **150**, the PWM controller **130** (e.g., an IC chip) undergoes a startup process. During the startup process, if the circuit **220** for undervoltage-lockout (UVLO) detects that the voltage **145** at the terminal **144** (e.g., a VDD pin) has not reached the startup threshold, the circuit **220** for undervoltage-lockout (UVLO) outputs a signal **223** to an input terminal of the charging controller **210**, and in response, the charging controller **210** generates a signal **211** at the output terminal to turn on the transistor **240** (e.g., a transistor M1) and to raise the voltage **145** at the terminal **144** (e.g., a VDD pin). If the circuit **220** for undervoltage-lockout (UVLO) detects that the voltage **145** at the terminal **144** (e.g., a VDD pin) has risen above the startup threshold, the circuit **220** for undervoltage-lockout (UVLO) outputs the signal **223** to the input terminal of the charging controller **210**, and in response, the charging controller **210** generates the signal **211** at the output terminal to turn off the transistor **240** (e.g., a transistor M1) in order to disconnect the charging circuit inside the controller **130** and starts normal operation for the power converter **100**.

(16) FIG. 3 is a simplified diagram showing another conventional power converter. The power converter **300** (e.g., an AC-DC power supply) includes a primary winding **302**, a secondary winding **304**, an auxiliary winding **306**, a transistor **310** (e.g., a transistor M1), a rectifier **320** (e.g., a bridge rectifier), a filter **322** for electromagnetic interference (EMI), an optocoupler **328**, a pulse-width-modulation (PWM) controller **330** (e.g., an IC chip), resistors **312**, **314**, **360**, **364**, **384**, **386**, **388** and **396**, capacitors **318**, **356**, **358**, **368**, **378** and **394**, diodes **362**, **370** and **376**, an error amplifier **398** (e.g., TL431), and an inductor **316**. The pulse-width-modulation (PWM) controller **330** (e.g., an IC chip) includes a terminal **332** (e.g., a SW pin), a terminal **334** (e.g., a V10 pin), a terminal **336** (e.g., an FB pin), a terminal **338** (e.g., a GND pin), a terminal **340** (e.g., a HV pin), a terminal **342** (e.g., a DEM pin), a terminal **344** (e.g., a GATE pin), a terminal **346** (e.g., a CS pin), and a terminal **348** (e.g., a VDD pin). Additionally, the optocoupler **328** includes a light emitting diode **392** and a photosensitive detector **390**. Moreover, the terminal **336** (e.g., an FB pin) is used to receive a feedback signal **337** that represents an output voltage **381** (e.g., V.sub.out) of the power converter **300**, and the terminal **340** (e.g., a HV pin) is used to receive a voltage **341**.

(17) As shown in FIG. 3, the EMI filter **322** receives an alternating current (AC) voltage **350**. Two output terminals of the EMI filter **322** are connected to two input terminals of the rectifier **320** (e.g., a bridge rectifier) respectively. The rectifier **320** (e.g., a bridge rectifier) includes four diodes (e.g., diodes D1, D2, D3, and D4). One output terminal of the rectifier **320** (e.g., a bridge rectifier) is connected to one plate of the capacitor **356** (e.g., a capacitor C1) that is biased to a ground voltage on the primary side, and another output terminal of the rectifier **320** (e.g., a bridge rectifier) is connected to another plate of the capacitor **356** (e.g., a capacitor C1) that is connected to one

terminal of the resistor **360** (e.g., a resistor R1), one terminal of the capacitor **358** (e.g., a capacitor C2), and one terminal of the primary winding **302**. Another terminal of the resistor **360** (e.g., a resistor R1) and another terminal of the capacitor **358** (e.g., a capacitor C2) both are connected to a cathode of the diode **362** (e.g., a diode D5). An anode of the diode **362** (e.g., a diode D5) is connected to a drain terminal of the transistor **310** (e.g., a transistor M1) and another terminal of the primary winding **302**.

(18) A gate terminal of the transistor **310** (e.g., a transistor M1) receives a drive signal **345** from the terminal **344** (e.g., a GATE pin) of the PWM controller **330** (e.g., an IC chip), and a source terminal of the transistor **310** (e.g., a transistor M1) is connected to the terminal **346** (e.g., a CS pin) of the PWM controller **330** (e.g., an IC chip) and also connected to one terminal of the resistor **364** (e.g., a resistor R4). Another terminal of the resistor **364** (e.g., a resistor R4) is biased to the ground voltage on the primary side. The terminal **348** (e.g., a VDD pin) of the pulse-width-modulation (PWM) controller **330** (e.g., an IC chip) is connected to one terminal of the inductor **316** (e.g., an inductor L), one plate of the capacitor **368** (e.g., a capacitor C3) and a cathode of the diode **370** (e.g., a diode D7). Another terminal of the inductor **316** (e.g., an inductor L) is connected to the terminal **332** (e.g., a SW pin) of the PWM controller **330** (e.g., an IC chip). Another plate of the capacitor **368** (e.g., a capacitor C3) is biased to the ground voltage on the primary side. An anode of the diode **370** (e.g., a diode D7) is connected to one terminal of the resistor **312** (e.g., a resistor R2) and one terminal of the auxiliary winding **306** that corresponds to a voltage **307**. Another terminal of the auxiliary winding **306** is biased to the ground voltage on the primary side. Another terminal of the resistor **312** (e.g., a resistor R2) is connected to the terminal **342** (e.g., a DEM pin) of the PWM controller **330** (e.g., an IC chip) and one terminal of the resistor **314** (e.g., a resistor R3). Another terminal of the resistor **314** (e.g., a resistor R3) is biased to the ground voltage on the primary side. The terminal **334** (e.g., a V10 pin) of the PWM controller **330** (e.g., an IC chip) is connected to one terminal of the capacitor **318** (e.g., a capacitor C5), and another terminal of the capacitor **318** (e.g., a capacitor C5) is biased to the ground voltage on the primary side.

(19) One terminal of the secondary winding **304** is connected to an anode of the diode **376** (e.g., a diode D6), and a cathode of the diode **376** (e.g., a diode D6) is connected to an output terminal **380** and one terminal of the capacitor **378** (e.g., a capacitor C4). Another terminal of the secondary winding **304** is connected to an output terminal **382** and another terminal of the capacitor **378** (e.g., a capacitor C4). The output terminal **380** provides the output voltage **381** (e.g., V.sub.out) and an output current **383** (e.g., I.sub.out), and the output terminal **382** is biased to a ground voltage on the secondary side.

(20) The output terminal **380** is connected to one terminal of the resistor **384** (e.g., a resistor R5), and another terminal of the resistor **384** (e.g., a resistor R5) is connected to one terminal of the resistor **396** (e.g., a resistor R6), one terminal of the resistor **386** (e.g., a resistor R7), and an input terminal of the error amplifier **398**. Another terminal of the resistor **396** (e.g., a resistor R6) is connected to one terminal of the capacitor **394** (e.g., a capacitor C5), which includes another terminal connected to a cathode of the light emitting diode **392** (e.g., a light emitting diode D8) and connected to an output terminal of the error amplifier **398**. Another terminal of the resistor **386** (e.g., R7) is biased to the ground voltage on the secondary side, and another input terminal of the error amplifier **398** is biased to a reference voltage **397**. An anode of the light emitting diode **392** (e.g., a light emitting diode D8) is connected to one terminal of the resistor **388** (e.g., a resistor R4), and another terminal of the resistor **388** (e.g., a resistor R4) is connected to the output terminal **380**. The terminal **338** (e.g., a GND pin) of the PWM controller **330** (e.g., an IC chip) is biased to the ground voltage on the primary side.

(21) As shown in FIG. 3, after the EMI filter **322** receives the alternating current (AC) voltage **350**, the voltage **341** is provided to the terminal **340** (e.g., a HV pin) of the PWM controller **330** (e.g., an IC chip) so that the capacitor **368** (e.g., a capacitor C3) is charged through the terminal **348** (e.g., a VDD pin). When an internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) does

not operate, the terminal **348** (e.g., a VDD pin) is used to raise a voltage **335** at the terminal **334** (e.g., a V10 pin) of the PWM controller **330** (e.g., an IC chip) by charging the capacitor **318** (e.g., a capacitor C5) through the external inductor **316** (e.g., an inductor L), the terminal **332** (e.g., a SW pin) and an internal Schottky diode of the PWM controller **330** (e.g., an IC chip). If the voltage **335** at the terminal **334** (e.g., a V10 pin) rises above a startup threshold, the PWM controller **330** (e.g., an IC chip) disconnects the charging circuit inside the controller **330** that is used to charge the capacitor **368** (e.g., a capacitor C3) by the voltage **341** at the terminal **340** (e.g., a HV pin), and the PWM controller **330** (e.g., an IC chip) also starts normal operation for the power converter **300**.
 (22) Under the normal operation, the output voltage **381** (e.g., V.sub.out) rises so that the voltage **307** at one terminal of the auxiliary winding **306** also rises. The auxiliary winding **306**, together with the diode **370** (e.g., a diode D7) and the capacitor **368** (e.g., a capacitor C3), provides power supply, to the terminal **348** (e.g., a VDD pin). A voltage **349** at the terminal **348** (e.g., a VDD pin) of the PWM controller **330** (e.g., an IC chip) is determined as follows:

$$(23) \quad V_{DD} \approx \frac{N_{aux}}{N_s} \times V_{out} \quad (\text{Equation 2})$$

where V.sub.DD represents the voltage **349** at the terminal **348** (e.g., a VDD pin), and V.sub.out represents the output voltage **381** at the output terminal **380**. Additionally, N.sub.aux represents the number of turns of the auxiliary winding **306**, and N.sub.s represents the number of turns of the secondary winding **304**.

(24) As shown by Equation 2, the higher the output voltage **381** (e.g., V.sub.out), the higher the voltage **349** (e.g., V.sub.DD), and the lower the output voltage **381** (e.g., V.sub.out), the lower the voltage **349** (e.g., V.sub.DD). When the output voltage **381** (e.g., V.sub.out) is low, the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) uses the voltage **349** (e.g., V.sub.DD) to generate the voltage **335** at the terminal **334** (e.g., a V10 pin), where the voltage **335** is stable and higher than the voltage **349** (e.g., V.sub.DD). The voltage **335** at the terminal **334** (e.g., a V10 pin) is used to provide stable power supply to the PWM controller **330** (e.g., an IC chip), even though the output voltage **381** (e.g., V.sub.out) varies widely.

(25) If the voltage **349** (e.g., V.sub.DD) is lower than the reference voltage (e.g., V.sub.reg), the voltage **335** at the terminal **334** (e.g., a V10 pin) is generated by the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) using the voltage **349** (e.g., V.sub.DD), and the generated voltage **335** is larger than the voltage **349** (e.g., V.sub.DD) and is equal to the reference voltage (e.g., V.sub.reg). If the voltage **349** (e.g., V.sub.DD) is higher than the reference voltage (e.g., V.sub.reg), the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) is idle, and the voltage **335** at the terminal **334** (e.g., a V10 pin) is approximately equal to the voltage **349** (e.g., V.sub.DD).

(26) Hence it is highly desirable to improve the techniques related to pulse-width-modulation (PWM) controllers.

3. BRIEF SUMMARY OF THE INVENTION

(27) Certain embodiments of the present invention are directed to circuits. More particularly, some embodiments of the invention provide systems and methods for providing power to pulse-width-modulation controllers of power converters. Merely by way of example, some embodiments of the invention have been applied to power converters during normal operation. But it would be recognized that the invention has a much broader range of applicability.

(28) According to some embodiments, a controller for a power converter includes: a first terminal configured to receive a first voltage; a second terminal connected to a capacitor and biased to a second voltage; a voltage detector configured to receive the second voltage from the second terminal and generate a detection signal based at least in part on the second voltage; a charging controller configured to receive the detection signal and generate a first control signal based at least in part on the detection signal; and a charging current generator configured to receive the first voltage from the first terminal and receive the first control signal from the charging controller;

wherein the voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the charging controller is further configured to: in response to the detection signal indicating that the second voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a charging current is to be generated; wherein the charging current generator is further configured to: in response to the first control signal indicating that the charging current is to be generated, generate the charging current that flows through the second terminal to charge the capacitor associated with the second voltage.

(29) According to certain embodiments, a controller for a power converter includes: a first terminal configured to receive a first voltage; a second terminal connected to a first capacitor and biased to a second voltage; a third terminal connected to a second capacitor and biased to a third voltage; a first voltage detector configured to receive the second voltage from the second terminal and generate a first detection signal based at least in part on the second voltage; a first charging controller configured to receive the first detection signal and generate a first control signal based at least in part on the first detection signal; and a charging current generator configured to receive the first voltage from the first terminal, the charging current generator including a first current generator and a second current generator, the first current generator being configured to receive the first control signal from the first charging controller; wherein the first voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the first charging controller is further configured to: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a first charging current is to be generated; wherein the first current generator is further configured to: in response to the first control signal indicating that the first charging current is to be generated, generate the first charging current that flows through the third terminal to charge the second capacitor associated with the third voltage.

(30) According to some embodiments, a controller for a power converter includes: a first terminal configured to receive a first voltage; a second terminal connected to a first capacitor and biased to a second voltage; a third terminal connected to a second capacitor and biased to a third voltage; a first voltage detector configured to receive the second voltage from the second terminal and generate a first detection signal based at least in part on the second voltage; a charging controller configured to receive the first detection signal and generate a first control signal based at least in part on the first detection signal; and a charging current generator configured to receive the first voltage from the first terminal and configured to receive the first control signal from the charging controller; wherein the first voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the charging controller is further configured to: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a charging current is to be generated; wherein the current generator is further configured to: in response to the first control signal indicating that the charging current is to be generated, generate the charging current that flows through the third terminal to charge the second capacitor associated with the third voltage.

(31) According to certain embodiments, a method for a power converter includes: receiving a first voltage; receiving a second voltage associated with a capacitor; generating a detection signal based at least in part on the second voltage; receiving the detection signal; generating a first control signal based at least in part on the detection signal; receiving the first control signal; and processing the first control signal; wherein the generating a detection signal based at least in part on the second

voltage includes: detecting that the second voltage has decreased to a first predetermined threshold; and generating the detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the generating a first control signal based at least in part on the detection signal includes: in response to the detection signal indicating that the second voltage has decreased to the first predetermined threshold, generating the first control signal indicating that a charging current is to be generated; wherein the processing the first control signal includes: in response to the first control signal indicating that the charging current is to be generated, generating the charging current to charge the capacitor associated with the second voltage.

(32) According to some embodiments, a method for a power converter includes: receiving a first voltage; receiving a second voltage associated with a first capacitor; receiving a third voltage associated with a second capacitor; generating a first detection signal based at least in part on the second voltage; receiving the first detection signal; generating a first control signal based at least in part on the first detection signal; receiving the first control signal; and processing the first control signal; wherein the generating a first detection signal based at least in part on the second voltage includes: detecting that the second voltage has decreased to a first predetermined threshold; and generating the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the generating a first control signal based at least in part on the first detection signal includes: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generating the first control signal indicating that a first charging current is to be generated; wherein the processing the first control signal includes: in response to the first control signal indicating that the first charging current is to be generated, generating the first charging current to charge the second capacitor associated with the third voltage.

(33) According to certain embodiments, a method for a power converter includes: receiving a first voltage; receiving a second voltage associated with a first capacitor; receiving a third voltage associated with a second capacitor; generating a first detection signal based at least in part on the second voltage; receiving the first detection signal; generating a first control signal based at least in part on the first detection signal; receiving the first control signal; and processing the first control signal; wherein the generating a first detection signal based at least in part on the second voltage includes: detecting that the second voltage has decreased to a first predetermined threshold; and generating the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the generating a first control signal based at least in part on the first detection signal includes: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generating the first control signal indicating that a charging current is to be generated; wherein the processing the first control signal includes: in response to the first control signal indicating that the charging current is to be generated, generating the charging current to charge the second capacitor associated with the third voltage.

(34) Depending upon embodiment, one or more benefits may be achieved. These benefits and various additional objects, features and advantages of the present invention can be fully appreciated with reference to the detailed description and accompanying drawings that follow.

Description

4. BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a simplified diagram showing a conventional power converter.

(2) FIG. 2 is a simplified diagram showing certain conventional components of the PWM controller of the power converter as shown in FIG. 1.

(3) FIG. 3 is a simplified diagram showing another conventional power converter.

- (4) FIG. 4 shows simplified timing diagrams under certain dynamic-load conditions for the power converter as shown in FIG. 1 according to certain embodiments.
- (5) FIG. 5 shows simplified timing diagrams under no-load condition for the power converter as shown in FIG. 1 according to some embodiments.
- (6) FIG. 6 is a simplified diagram showing a power converter according to certain embodiments of the present invention.
- (7) FIG. 7 is a simplified diagram showing a power converter according to some embodiments of the present invention.
- (8) FIG. 8 is a simplified diagram showing certain components of the PWM controller of the power converter as shown in FIG. 6 and/or certain components of the PWM controller of the power converter as shown in FIG. 7 according to some embodiments of the present invention.
- (9) FIG. 9 is a simplified diagram showing certain components of the PWM controller of the power converter as shown in FIG. 6 according to some embodiments of the present invention.
- (10) FIG. 10 shows simplified timing diagrams under certain dynamic-load and/or no-load conditions for the power converter that includes the PWM controller as shown in FIG. 6 and FIG. 9 according to some embodiments.
- (11) FIG. 11 shows simplified timing diagrams under certain dynamic-load conditions for the power converter as shown in FIG. 3 according to certain embodiments.
- (12) FIG. 12 shows simplified timing diagrams under no-load condition for the power converter as shown in FIG. 3 according to certain embodiments.
- (13) FIG. 13 is a simplified diagram showing certain components of the PWM controller of the power converter as shown in FIG. 7 according to some embodiments of the present invention.
- (14) FIG. 14 shows simplified timing diagrams under certain dynamic-load and/or no-load conditions for the power converter that includes the PWM controller as shown in FIG. 7 and FIG. 13 according to some embodiments of the present invention.
- (15) FIG. 15 is a simplified diagram showing certain components of the PWM controller of the power converter as shown in FIG. 7 according to some embodiments of the present invention.
- (16) FIG. 16 shows a simplified diagram showing certain charging current as a function of certain voltage for the power converter that includes the PWM controller as shown in FIG. 7 and FIG. 15 according to some embodiments of the present invention.
- (17) FIG. 17 is a simplified diagram showing certain components of the PWM controller of the power converter as shown in FIG. 7 according to some embodiments of the present invention.

5. DETAILED DESCRIPTION OF THE INVENTION

- (18) Certain embodiments of the present invention are directed to circuits. More particularly, some embodiments of the invention provide systems and methods for providing power to pulse-width-modulation controllers of power converters. Merely by way of example, some embodiments of the invention have been applied to power converters during normal operation. But it would be recognized that the invention has a much broader range of applicability.
- (19) As shown in FIG. 1 and FIG. 2, after the power converter **100** starts the normal operation, the auxiliary winding **106** is used to supply power to the terminal **144** (e.g., a VDD pin) of the PWM controller **130** (e.g., an IC chip) according to some embodiments. For example, under certain dynamic-load and/or no-load conditions, the PWM controller **130** (e.g., an IC chip) keeps the gate signal **147** at a logic low level for multiple clock cycles due to the adjustment of the feedback signal **135**. As an example, during these multiple clock cycles, the auxiliary winding **106** does not provide additional power, and the voltage **145** at the terminal **144** (e.g., a VDD pin) is supported by the capacitor **168** (e.g., a capacitor C1). According to certain embodiments, with the discharge of the capacitor **168** (e.g., a capacitor C1), the voltage **145** at the terminal **144** (e.g., a VDD pin) decreases with time. For example, if the voltage **145** at the terminal **144** (e.g., a VDD pin) drops below a predetermined threshold, the output voltage **181** (e.g., V_{sub.out}) drifts up. As an example, under the low temperature and/or a harsh working condition, with the discharge of the capacitor

168 (e.g., a capacitor **C1**), the voltage **145** at the terminal **144** (e.g., a VDD pin) decreases more quickly due to the decrease in capacitance of the capacitor **168** (e.g., a capacitor **C1**), causing insufficient power supply for the PWM controller **130** (e.g., an IC chip) and also triggering undervoltage protection.

(20) FIG. 4 shows simplified timing diagrams under certain dynamic-load conditions for the power converter **100** as shown in FIG. 1 according to certain embodiments. The waveform **410** represents the output current **183** (e.g., $I_{sub.out}$) as a function of time, the waveform **420** represents the output voltage **181** (e.g., $V_{sub.out}$) as a function of time, the waveform **430** represents the feedback signal **135** as a function of time, the waveform **440** represents the drive signal **147** as a function of time, and the waveform **450** represents the voltage **145** (e.g., $V_{sub.DD}$) as a function of time.

(21) In some embodiments, before time $t_{sub.1}$, the output current **183** (e.g., $I_{sub.out}$) is at a current level **412**, and within each clock cycle, the drive signal **147** changes between a logic high level and a logic low level. For example, at time $t_{sub.1}$, the output current **183** (e.g., $I_{sub.out}$) decreases from the current level **412** to a current level **414**. As an example, from time $t_{sub.1}$ to time $t_{sub.2}$, the output current **183** (e.g., $I_{sub.out}$) remains at the current level **414**. For example, at time $t_{sub.2}$, the output current **183** (e.g., $I_{sub.out}$) increases from the current level **414** to the current level **412**. As an example, after time $t_{sub.2}$, the output current **183** (e.g., $I_{sub.out}$) remains at the current level **412**. In some examples, the current level **412** is higher than the current level **414**. For example, if the output current **183** (e.g., $I_{sub.out}$) is at the current level **412**, the power converter **100** is under a high-load condition. As an example, if the output current **183** (e.g., $I_{sub.out}$) is at the current level **414**, the power converter **100** is under a light-load condition. In certain embodiments, from time $t_{sub.1}$ to time $t_{sub.2}$, the power converter **100** is under the light-load condition. For example, under the light-load condition, the output voltage **181** (e.g., $V_{sub.out}$) overshoots as shown by the waveform **420**, the feedback signal **135** drops as shown by the waveform **430**, the drive signal **147** remains at the logic low level during multiple clock cycles as shown by the waveform **440**, and the voltage **145** (e.g., $V_{sub.DD}$) decreases gradually from time $t_{sub.1}$ to time $t_{sub.2}$ as shown by the waveform **450**.

(22) FIG. 5 shows simplified timing diagrams under no-load condition for the power converter **100** as shown in FIG. 1 according to some embodiments. The waveform **530** represents the feedback signal **135** as a function of time, the waveform **540** represents the drive signal **147** as a function of time, and the waveform **550** represents the voltage **145** (e.g., VDD) as a function of time.

(23) According to certain embodiments, under no-load condition, the output current **183** (e.g., $I_{sub.out}$) is equal to zero in magnitude. For example, at time $t_{sub.10}$, the feedback signal **135** becomes larger than a signal threshold **532** (e.g., FB_Burst_H), and at time $t_{sub.11}$, the feedback signal **135** becomes smaller than a signal threshold **534** (e.g., FB_Burst_L) as shown by the waveform **530**. As an example, at time $t_{sub.12}$, the feedback signal **135** becomes larger than the signal threshold **532** (e.g., FB_Burst_H), and at time $t_{sub.13}$, the feedback signal **135** becomes smaller than the signal threshold **534** (e.g., FB_Burst_L) as shown by the waveform **530**. For example, the signal threshold **532** (e.g., FB_Burst_H) is larger than the signal threshold **534** (e.g., FB_Burst_L) in magnitude.

(24) In some embodiments, from time $t_{sub.10}$ to time $t_{sub.11}$, the drive signal **147** changes between a logic high level and a logic low level within each clock cycle as shown by the waveform **540**, and the voltage **145** (e.g., $V_{sub.DD}$) increases as shown by the waveform **550**. In certain embodiments, time $t_{sub.11}$ to time $t_{sub.12}$, the drive signal **147** remains at the logic low level during multiple clock cycles as shown by the waveform **540**, and the voltage **145** (e.g., $V_{sub.DD}$) decreases gradually as shown by the waveform **550**. In some embodiments, from time $t_{sub.12}$ to time $t_{sub.13}$, the drive signal **147** changes between the logic high level and the logic low level within each clock cycle as shown by the waveform **540**, and the voltage **145** (e.g., $V_{sub.DD}$) increases as shown by the waveform **550**.

(25) As shown in FIG. 4 and FIG. 5, under certain dynamic-load and/or no-load conditions, when the drive signal **147** remains at the logic low level during multiple clock cycles, the voltage **145** (e.g., V.sub.DD) decreases gradually according to certain embodiments. For example, if the voltage **145** (e.g., V.sub.DD) becomes too small in magnitude, the undervoltage protection is triggered for the PWM controller **130** (e.g., an IC chip) because of insufficient power supply.

(26) FIG. 6 is a simplified diagram showing a power converter according to certain embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The power converter **600** (e.g., an AC-DC power supply) includes a primary winding **602**, a secondary winding **604**, an auxiliary winding **606**, a transistor **610** (e.g., a transistor M1), a rectifier **620** (e.g., a bridge rectifier), a filter **622** for electromagnetic interference (EMI), diodes **624**, **626**, **662**, **670** and **676**, an optocoupler **628**, a pulse-width-modulation (PWM) controller **630** (e.g., an IC chip), resistors **652**, **654**, **660**, **664**, **666**, **672**, **674**, **684**, **686**, **688** and **696**, capacitors **656**, **658**, **668**, **678** and **694**, and an error amplifier **698**, where the pulse-width-modulation (PWM) controller **630** (e.g., an IC chip) includes the PWM controller **800** as shown in FIG. 8 and/or the PWM controller **630** as shown in FIG. 9. For example, the pulse-width-modulation (PWM) controller **630** (e.g., an IC chip) includes a terminal **632** (e.g., a PRT pin), a terminal **634** (e.g., an FB pin), a terminal **636** (e.g., a CS pin), a terminal **638** (e.g., a GND pin), a terminal **640** (e.g., a HV pin), a terminal **642** (e.g., an NC pin), a terminal **644** (e.g., a VDD pin), and a terminal **646** (e.g., a GATE pin). As an example, the optocoupler **628** includes a light emitting diode **692** and a photosensitive detector **690**. For example, the terminal **634** (e.g., an FB pin) is used to receive a feedback signal **635** that represents an output voltage **681** (e.g., V.sub.out) of the power converter **600**, and the terminal **640** (e.g., a HV pin) is used to receive a voltage **641**. As an example, the error amplifier **698** includes TL431.

(27) As shown in FIG. 6, the EMI filter **622** receives an alternating current (AC) voltage **650** according to some embodiments. For example, one output terminal of the EMI filter **622** is connected to an anode of the diode **624** (e.g., a diode D1), and another output terminal of the EMI filter **622** is connected to an anode of the diode **626** (e.g., a diode D2). As an example, a cathode of the diode **624** (e.g., a diode D1) and a cathode of the diode **626** (e.g., a diode D2) both are connected to one terminal of the resistor **652** (e.g., a resistor R2), which includes another terminal that is connected to one terminal of the resistor **654** (e.g., a resistor R3). For example, another terminal of the resistor **654** (e.g., a resistor R3) is connected to the terminal **640** (e.g., a HV pin) of the PWM controller **630**.

(28) According to certain embodiments, the anode of the diode **624** (e.g., a diode D1) is connected to an input terminal of the rectifier **620** (e.g., a bridge rectifier), and the anode of the diode **626** (e.g., a diode D2) is connected to another input terminal of the rectifier **620** (e.g., a bridge rectifier). For example, the rectifier **620** (e.g., a bridge rectifier) includes four diodes (e.g., diodes D3, D4, D5, and D6). As an example, one output terminal of the rectifier **620** (e.g., a bridge rectifier) is connected to one plate of the capacitor **656** (e.g., a capacitor C2) that is biased to a ground voltage on the primary side, and another output terminal of the rectifier **620** (e.g., a bridge rectifier) is connected to another plate of the capacitor **656** (e.g., a capacitor C2) that is connected to one terminal of the resistor **660** (e.g., a resistor R1), one terminal of the capacitor **658** (e.g., a capacitor C3), and one terminal of the primary winding **602**. For example, another terminal of the resistor **660** (e.g., a resistor R1) and another terminal of the capacitor **658** (e.g., a capacitor C3) both are connected to a cathode of the diode **662** (e.g., a diode D7). As an example, an anode of the diode **662** (e.g., a diode D7) is connected to a drain terminal of the transistor **610** (e.g., a transistor M1) and another terminal of the primary winding **602**.

(29) In some embodiments, a gate terminal of the transistor **610** (e.g., a transistor M1) receives a drive signal **647** from the terminal **646** (e.g., a GATE pin) of the PWM controller **630** (e.g., an IC chip), and a source terminal of the transistor **610** (e.g., a transistor M1) is connected to the terminal

636 (e.g., a CS pin) of the PWM controller **630** (e.g., an IC chip) and also connected to one terminal of the resistor **664** (e.g., a resistor R7). For example, another terminal of the resistor **664** (e.g., a resistor R7) is biased to the ground voltage on the primary side. As an example, the terminal **644** (e.g., a VDD pin) of the pulse-width-modulation (PWM) controller **630** (e.g., an IC chip) is connected to one terminal of the resistor **666** (e.g., a resistor R6), and another terminal of the resistor **666** (e.g., a resistor R6) is connected to one plate of the capacitor **668** (e.g., a capacitor C1) and a cathode of the diode **670** (e.g., a diode D8). For example, the terminal **644** (e.g., a VDD pin) is at a voltage **645**. As an example, another plate of the capacitor **668** (e.g., a capacitor C1) is biased to the ground voltage on the primary side. For example, an anode of the diode **670** (e.g., a diode D8) is connected to one terminal of the auxiliary winding **606**, and another terminal of the auxiliary winding **606** is biased to the ground voltage on the primary side.

(30) In certain embodiments, the anode of the diode **670** (e.g., a diode D8) is connected to one terminal of the resistor **672** (e.g., a resistor R4), and another terminal of the resistor **672** (e.g., a resistor R4) is connected to the terminal **632** (e.g., a PRT pin) of the PWM controller **630** (e.g., an IC chip) and also connected to one terminal of the resistor **674** (e.g., a resistor R5). For example, another terminal of the resistor **674** (e.g., a resistor R5) is biased to the ground voltage on the primary side. As an example, one terminal of the secondary winding **604** is connected to an anode of the diode **676** (e.g., a diode D9), and a cathode of the diode **676** (e.g., a diode D9) is connected to an output terminal **680** and one terminal of the capacitor **678** (e.g., a capacitor C4). For example, another terminal of the secondary winding **604** is connected to an output terminal **682** and another terminal of the capacitor **678** (e.g., a capacitor C4). As an example, the output terminal **680** provides the output voltage **681** (e.g., $V_{sub.out}$) and an output current **683** (e.g., $I_{sub.out}$), and the output terminal **682** is biased to a ground voltage on the secondary side.

(31) According to some embodiments, the output terminal **680** is connected to one terminal of the resistor **684** (e.g., a resistor R10), and another terminal of the resistor **684** (e.g., a resistor R10) is connected to one terminal of the resistor **696** (e.g., a resistor R9), one terminal of the resistor **686** (e.g., a resistor R11), and an input terminal of the error amplifier **698**. For example, another terminal of the resistor **696** (e.g., a resistor R9) is connected to one terminal of the capacitor **694** (e.g., a capacitor C5), which includes another terminal connected to a cathode of the light emitting diode **692** (e.g., a light emitting diode D10) and connected to an output terminal of the error amplifier **698**. As an example, another terminal of the resistor **686** (e.g., R11) is biased to the ground voltage on the secondary side, and another input terminal of the error amplifier **698** is biased to a reference voltage **697**. For example, an anode of the light emitting diode **692** (e.g., a light emitting diode D10) is connected to one terminal of the resistor **688** (e.g., a resistor R8), and another terminal of the resistor **688** (e.g., a resistor R8) is connected to the output terminal **680**. As an example, the terminal **638** (e.g., a GND pin) of the PWM controller **630** (e.g., an IC chip) is biased to the ground voltage on the primary side. According to certain embodiments, under the normal operation, when the output voltage **681** (e.g., $V_{sub.out}$) rises, the auxiliary winding **606**, together with the diode **670** (e.g., a diode D8), the capacitor **668** (e.g., a capacitor C1) and the resistor **666** (e.g., a resistor R6), raise the voltage **645** at the terminal **644** (e.g., a VDD pin).

(32) FIG. 7 is a simplified diagram showing a power converter according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The power converter **700** (e.g., an AC-DC power supply) includes a primary winding **702**, a secondary winding **704**, an auxiliary winding **706**, a transistor **710** (e.g., a transistor M1), a rectifier **720** (e.g., a bridge rectifier), a filter **722** for electromagnetic interference (EMI), an optocoupler **728**, a pulse-width-modulation (PWM) controller **730** (e.g., an IC chip), resistors **712**, **714**, **760**, **764**, **784**, **786**, **788** and **796**, capacitors **718**, **756**, **758**, **768**, **778** and **794**, diodes **762**, **770** and **776**, an error amplifier **798**, and an inductor **716**, where the pulse-width-modulation (PWM) controller **730** (e.g., an IC chip) includes the PWM controller **800** as shown in FIG. 8, the PWM

controller **730** as shown in FIG. **13**, the PWM controller **730** as shown in FIG. **15**, and/or the PWM controller **730** as shown in FIG. **17**. For example, the pulse-width-modulation (PWM) controller **730** (e.g., an IC chip) includes a terminal **732** (e.g., a SW pin), a terminal **734** (e.g., a V₁₀ pin), a terminal **736** (e.g., an FB pin), a terminal **738** (e.g., a GND pin), a terminal **740** (e.g., a HV pin), a terminal **742** (e.g., a DEM pin), a terminal **744** (e.g., a GATE pin), a terminal **746** (e.g., a CS pin), and a terminal **748** (e.g., a VDD pin). As an example, the optocoupler **728** includes a light emitting diode **792** and a photosensitive detector **790**. For example, the terminal **736** (e.g., an FB pin) is used to receive a feedback signal **737** that represents an output voltage **781** (e.g., V_{sub.out}) of the power converter **700**, and the terminal **740** (e.g., a HV pin) is used to receive a voltage **741**. As an example, the error amplifier **798** includes TL**431**.

(33) As shown in FIG. **7**, the EMI filter **722** receives an alternating current (AC) voltage **750** according to some embodiments. For example, two output terminals of the EMI filter **722** are connected to two input terminals of the rectifier **720** (e.g., a bridge rectifier) respectively. As an example, the rectifier **720** (e.g., a bridge rectifier) includes four diodes (e.g., diodes **D1**, **D2**, **D3**, and **D4**). For example, one output terminal of the rectifier **720** (e.g., a bridge rectifier) is connected to one plate of the capacitor **756** (e.g., a capacitor **C1**) that is biased to a ground voltage on the primary side, and another output terminal of the rectifier **720** (e.g., a bridge rectifier) is connected to another plate of the capacitor **756** (e.g., a capacitor **C1**) that is connected to one terminal of the resistor **760** (e.g. a resistor **R1**), one terminal of the capacitor **758** (e.g., a capacitor **C2**), and one terminal of the primary winding **702**. As an example, another terminal of the resistor **760** (e.g. a resistor **R1**) and another terminal of the capacitor **758** (e.g., a capacitor **C2**) both are connected to a cathode of the diode **762** (e.g., a diode **D5**). For example, an anode of the diode **762** (e.g., a diode **D5**) is connected to a drain terminal of the transistor **710** (e.g., a transistor **M1**) and another terminal of the primary winding **702**.

(34) According to certain embodiments, a gate terminal of the transistor **710** (e.g., a transistor **M1**) receives a drive signal **745** from the terminal **744** (e.g., a GATE pin) of the PWM controller **730** (e.g., an IC chip), and a source terminal of the transistor **710** (e.g., a transistor **M1**) is connected to the terminal **746** (e.g., a CS pin) of the PWM controller **730** (e.g., an IC chip) and also connected to one terminal of the resistor **764** (e.g., a resistor **R4**). For example, another terminal of the resistor **764** (e.g., a resistor **R4**) is biased to the ground voltage on the primary side. As an example, the terminal **748** (e.g., a VDD pin) of the pulse-width-modulation (PWM) controller **730** (e.g., an IC chip) is connected to one terminal of the inductor **716** (e.g., an inductor **L**), one plate of the capacitor **768** (e.g., a capacitor **C3**) and a cathode of the diode **770** (e.g., a diode **D7**). For example, the terminal **748** (e.g., a VDD pin) is at a voltage **749**. As an example, another terminal of the inductor **716** (e.g., an inductor **L**) is connected to the terminal **732** (e.g., a SW pin) of the PWM controller **730** (e.g., an IC chip). For example, another plate of the capacitor **768** (e.g., a capacitor **C3**) is biased to the ground voltage on the primary side. As an example, an anode of the diode **770** (e.g., a diode **D7**) is connected to one terminal of the resistor **712** (e.g., a resistor **R2**) and one terminal of the auxiliary winding **706** that corresponds to a voltage **707**. For example, another terminal of the auxiliary winding **706** is biased to the ground voltage on the primary side. As an example, another terminal of the resistor **712** (e.g., a resistor **R2**) is connected to the terminal **742** (e.g., a DEM pin) of the PWM controller **730** (e.g., an IC chip) and one terminal of the resistor **714** (e.g., a resistor **R3**). For example, another terminal of the resistor **714** (e.g., a resistor **R3**) is biased to the ground voltage on the primary side. As an example, the terminal **734** (e.g., a V₁₀ pin) of the PWM controller **730** (e.g., an IC chip) is connected to one terminal of the capacitor **718** (e.g., a capacitor **C5**), and another terminal of the capacitor **718** (e.g., a capacitor **C5**) is biased to the ground voltage on the primary side. For example, the terminal **734** (e.g., a V₁₀ pin) is at a voltage **735**.

(35) In some embodiments, one terminal of the secondary winding **704** is connected to an anode of the diode **776** (e.g., a diode **D6**), and a cathode of the diode **776** (e.g., a diode **D6**) is connected to

an output terminal **780** and one terminal of the capacitor **778** (e.g., a capacitor C4). For example, another terminal of the secondary winding **704** is connected to an output terminal **782** and another terminal of the capacitor **778** (e.g., a capacitor C4). As an example, the output terminal **780** provides the output voltage **781** (e.g., V.sub.out) and an output current **783** (e.g., I.sub.out), and the output terminal **782** is biased to a ground voltage on the secondary side.

(36) According to certain embodiments, the output terminal **780** is connected to one terminal of the resistor **784** (e.g., a resistor R5), and another terminal of the resistor **784** (e.g., a resistor R5) is connected to one terminal of the resistor **796** (e.g., a resistor R6), one terminal of the resistor **786** (e.g., a resistor R7), and an input terminal of the error amplifier **798**. For example, another terminal of the resistor **796** (e.g., a resistor R6) is connected to one terminal of the capacitor **794** (e.g., a capacitor C5), which includes another terminal connected to a cathode of the light emitting diode **792** (e.g., a light emitting diode D8) and connected to an output terminal of the error amplifier **798**. As an example, another terminal of the resistor **786** (e.g., R7) is biased to the ground voltage on the secondary side, and another input terminal of the error amplifier **798** is biased to a reference voltage **797**. For example, an anode of the light emitting diode **792** (e.g., a light emitting diode D8) is connected to one terminal of the resistor **788** (e.g., a resistor R4), and another terminal of the resistor **788** (e.g., a resistor R4) is connected to the output terminal **780**. As an example, the terminal **738** (e.g., a GND pin) of the PWM controller **730** (e.g., an IC chip) is biased to the ground voltage on the primary side. According to certain embodiments, under the normal operation, when the output voltage **781** (e.g., V.sub.out) rises, the auxiliary winding **706**, together with the diode **770** (e.g., a diode D7) and the capacitor **768** (e.g., a capacitor C3), raises the voltage **749** at the terminal **748** (e.g., a VDD pin).

(37) FIG. 8 is a simplified diagram showing certain components of the PWM controller **630** of the power converter **600** as shown in FIG. 6 and/or certain components of the PWM controller **730** of the power converter **700** as shown in FIG. 7 according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The PWM controller **800** (e.g., the PWM controller **630** and/or the PWM controller **730**) includes a charging circuit **810**, a voltage detection circuit **820**, and a charging controller **830**. Additionally, the PWM controller **800** (e.g., an IC chip) includes a terminal **870** (e.g., a pin), a terminal **880** (e.g., a pin) and a terminal **890** (e.g., a pin). In some examples, the PWM controller **800** (e.g., an IC chip) is the PWM controller **630** (e.g., an IC chip), the terminal **870** (e.g., a pin) is the terminal **640** (e.g., a HV pin), the terminal **880** (e.g., a pin) is the terminal **644** (e.g., a VDD pin), and the terminal **890** (e.g., a pin) is also the terminal **644** (e.g., a VDD pin), where the terminal **880** (e.g., a pin) and the terminal **890** (e.g., a pin) are the same terminal. In certain examples, the PWM controller **800** (e.g., an IC chip) is the PWM controller **730** (e.g., an IC chip), the terminal **870** (e.g., a pin) is the terminal **740** (e.g., a HV pin), the terminal **880** (e.g., a pin) is the terminal **748** (e.g., a VDD pin), and the terminal **890** (e.g., a pin) is the terminal **734** (e.g., a V10 pin).

(38) According to certain embodiments, the voltage detection circuit **820** receives a signal **882** from the terminal **880** (e.g., terminal **644** and/or the terminal **748**) and generates a signal **822** based at least in part on the signal **882**. In some examples, the charging controller **830** receives the signal **822** and generates a signal **832** based in part on the signal **822**. For example, the signal **832** is used to connect (e.g., turn on) the charging circuit **810** so that the charging circuit **810** is allowed to use a signal **872** from the terminal **870** (e.g., the terminal **640** and/or the terminal **740**) to charge the terminal **890** (e.g., the terminal **644** and/or the terminal **734**). As an example, the signal **832** is used to disconnect (e.g., turn off) the charging circuit **810** so that the charging circuit **810** is not allowed to use the signal **872** from the terminal **870** (e.g., the terminal **640** and/or the terminal **740**) to charge the terminal **890** (e.g., the terminal **644** and/or the terminal **734**). In certain examples, the charging circuit **810** receives the signal **832** from the charging controller **830** and the signal **872**

from the terminal **870** (e.g., the terminal **640** and/or the terminal **740**) and generates a signal **892** based at least in part on the signals **832** and **872**. For example, the signal **892** is received by the terminal **890** (e.g., the terminal **644** and/or the terminal **734**).

(39) According to some embodiments, the voltage detection circuit **820** detects whether the signal **882** (e.g., the voltage **645** and/or the voltage **749**) drops to a lower predetermined threshold (e.g., VDD_HV_ON) and whether the signal **882** (e.g., the voltage **645** and/or the voltage **749**) rises to a higher predetermined threshold (e.g., VDD_HV_OFF). For example, the higher predetermined threshold (e.g., VDD_HV_OFF) is larger than the lower predetermined threshold (e.g., VDD_HV_ON). As an example, the voltage detection circuit **820** detects whether the signal **882** (e.g., the voltage **645** and/or the voltage **749**) drops to the lower predetermined threshold (e.g., VDD_HV_ON) and whether the signal **882** (e.g., the voltage **645** and/or the voltage **749**) rises to the higher predetermined threshold (e.g., VDD_HV_OFF) under certain dynamic-load and/or no-load conditions of the power converter **600** and/or the power converter **700**.

(40) In certain examples, if the voltage detection circuit **820** determines that the signal **882** (e.g., the voltage **645** and/or the voltage **749**) drops to the lower predetermined threshold (e.g., VDD_HV_ON), the voltage detection circuit **820** generates the signal **822** (e.g., Char_v10), and in response to receiving the signal **822** (e.g., Char_v10), the charging controller **830** generates the signal **832** in order to connect (e.g., turn on) the charging circuit **810**. As an example, if the charging circuit **810** is connected (e.g., turned on), the charging circuit **810** uses the signal **872** (e.g., the voltage **641** and/or the voltage **741**) to generate the signal **892** and charge the terminal **890** (e.g., the terminal **644** and/or the terminal **734**). For example, when the terminal **890** (e.g., the terminal **644** and/or the terminal **734**) is being charged by the charging circuit **810**, the voltage **645** and/or the voltage **735** at the terminal **890** (e.g., the terminal **644** and/or the terminal **734**) is supported by the signal **892** from the charging circuit **810**, in addition to being supported by charges on the capacitor **668** and/or the capacitor **718**, in order to avoid triggering undervoltage protection in response to insufficient voltage **645** and/or the voltage **735** under certain dynamic-load and/or no-load conditions of the power converter **600** and/or the power converter **700** respectively.

(41) In some examples, if the voltage detection circuit **820** determines that the signal **882** (e.g., the voltage **645** and/or the voltage **749**) rises to the higher predetermined threshold (e.g., VDD_HV_OFF), the voltage detection circuit **820** generates the signal **822** (e.g., Char_v10), and in response to receiving the signal **822** (e.g., Char_v10), the charging controller **830** generates the signal **832** in order to disconnect (e.g., turn off) the charging circuit **810**. As an example, if the charging circuit **810** is disconnected (e.g., turned off), the charging circuit **810** does not charge the terminal **890** (e.g., the terminal **644** and/or the terminal **734**), in order to avoid extra power consumption by the charging circuit **810** and to improve power efficiency of the power converter **600** and/or the power converter **700** respectively.

(42) FIG. 9 is a simplified diagram showing certain components of the PWM controller **630** of the power converter **600** as shown in FIG. 6 according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The PWM controller **630** includes a charging circuit **910**, a voltage detection circuit **920**, a charging controller **930**, a circuit **980** for undervoltage-lockout (UVLO), and an internal control circuit **990**. Additionally, the PWM controller **630** (e.g., an IC chip) includes the terminal **640** (e.g., a HV pin) and the terminal **644** (e.g., a VDD pin). In certain examples, the charging circuit **810** includes a transistor **902** (e.g., a junction field-effect transistor), a resistor **904** (e.g., a resistor R1), and a transistor **906** (e.g., a metal-oxide-semiconductor field-effect transistor). In some examples, the charging circuit **910** is an example of the charging circuit **810**, the voltage detection circuit **920** is an example of the voltage detection circuit **820**, and the charging controller **930** is an example of the charging controller **830**.

(43) In certain embodiments, a drain terminal of the transistor **902** (e.g., JFET) is connected to the terminal **640** (e.g., a HV pin), the gate terminal of the transistor **902** (e.g., JFET) is biased to the ground voltage on the primary side of the power converter **600**, and a source terminal of the transistor **902** (e.g., JFET) is connected to one terminal of the resistor **904** (e.g., a resistor R1). For example, another terminal of the resistor **904** (e.g., a resistor R1) is connected to a source terminal of the transistor **906** (e.g., a transistor M2). As an example, a gate terminal of the transistor **906** (e.g., a transistor M2) receives a signal **932** (e.g., the signal **832**) from the charging controller **930** (e.g., the charging controller **830**). For example, a drain terminal of the transistor **906** (e.g., a transistor M2) is connected to the terminal **644** (e.g., a VDD pin).

(44) In some embodiments, an input terminal of the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) receives the voltage **645** from the terminal **644** (e.g., a VDD pin). For example, the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) generates a signal **922** (e.g., the signal **822**) and sends the signal **922** (e.g., the signal **822**) to the charging controller **930** (e.g., the charging controller **830**). As an example, the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) generates a signal **924** and sends the signal **924** to the circuit **980** for undervoltage-lockout (UVLO). In certain examples, an output terminal of the circuit **980** for undervoltage-lockout (UVLO) is connected to an input terminal of the internal control circuit **990**. As an example, the internal control circuit **990** includes various control circuits, and the internal control circuit **990** does not include the charging controller **930**. For example, the internal control circuit **990** includes one or more control circuits for one or more oscillators, one or more control circuits for one or more timers, one or more drivers, one or more logic control circuits, and/or one or more protection circuits.

(45) As shown in FIG. 6, after the EMI filter **622** receives the AC voltage **650**, the PWM controller **630** (e.g., an IC chip) undergoes a startup process according to certain embodiments. In some examples, during the startup process, if the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) determines that the voltage **645** has not reached a startup threshold (e.g., VDD_UVLO), the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) generates the signal **922** (e.g., the signal **822**), so that the charging controller **930** generates the signal **932** (e.g., the signal **832**) to turn on the transistor **906** (e.g., a transistor M2) and the charging circuit **910** (e.g., the charging circuit **810**) generates a charging current **992** (e.g., the signal **892**) to charge the capacitor **668** (e.g., a capacitor C1) and raise the voltage **645**, until the voltage **645** reaches the startup threshold (e.g., VDD_UVLO). In certain examples, before the voltage **645** reaches the startup threshold (e.g., VDD_UVLO), the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) generates the signal **924**, so that the circuit **980** for undervoltage-lockout (UVLO) does not generate the signal **921** that indicates the end of the startup process and causes the internal control circuit **990** to power on other components of the PWM controller **630** (e.g., an IC chip). For example, during the startup process, other components of the PWM controller **630** (e.g., an IC chip) remain powered off.

(46) According to some embodiments, if the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) determines that the voltage **645** has risen above the startup threshold (e.g., VDD_UVLO), the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) generates the signal **922** (e.g., the signal **822**), so that the charging controller **930** generates the signal **932** (e.g., the signal **832**) to turn off the transistor **906** (e.g., a transistor M2) and the charging circuit **910** (e.g., the charging circuit **810**) does not charge the capacitor **668** (e.g., a capacitor C1). In some examples, after the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) determines that the voltage **645** has risen above the startup threshold (e.g., VDD_UVLO), the startup process of the PWM controller **630** (e.g., an IC chip) ends and the normal operation for the power converter **600** starts. For example, at the beginning of the normal operation, the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) generates the signal **924**, so that the circuit **980** for undervoltage-lockout (UVLO) generates the signal **921** that indicates the end of the startup process

and causes the internal control circuit **990** to power on other components of the PWM controller **630** (e.g., an IC chip). As an example, during the normal operation, other components of the PWM controller **630** (e.g., an IC chip) remain powered on.

(47) According to certain embodiments, the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) also determines whether the voltage **645** drops to a lower predetermined threshold (e.g., VDD_HV_ON) and whether the voltage **645** rises to a higher predetermined threshold (e.g., VDD_HV_OFF). For example, the higher predetermined threshold (e.g., VDD_HV_OFF) is larger than the lower predetermined threshold (e.g., VDD_HV_ON), and the lower predetermined threshold (e.g., VDD_HV_ON) is larger than the startup threshold (e.g., VDD_UVLO).

(48) In certain embodiments, during the normal operation, if the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) determines that the voltage **645** drops to a lower predetermined threshold (e.g., VDD_HV_ON), the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) generates the signal **922** (e.g., the signal **822**), so that the charging controller **930** generates the signal **932** (e.g., the signal **832**) to turn on the transistor **906** (e.g., a transistor M2). For example, if the transistor **906** (e.g., a transistor M2) is turned on, the charging circuit **910** (e.g., the charging circuit **810**) generates a charging current **992** (e.g., the signal **892**) to flow through the terminal **644** (e.g., a VDD pin) and to charge the capacitor **668** (e.g., a capacitor C1) and raise the voltage **645**. As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **910** (e.g., the charging circuit **810**) generates the charging current **992** (e.g., the signal **892**) to flow through the terminal **644** (e.g., a VDD pin) and to charge the capacitor **668** (e.g., a capacitor C1) and raise the voltage **645**, in order to prevent insufficient power supply to the PWM controller **630** (e.g., an IC chip).

(49) In some embodiments, during the normal operation, if the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) determines that the voltage **645** rises to a higher predetermined threshold (e.g., VDD_HV_OFF), the voltage detection circuit **920** (e.g., the voltage detection circuit **820**) generates the signal **922** (e.g., the signal **822**), so that the charging controller **930** generates the signal **932** (e.g., the signal **832**) to turn off the transistor **906** (e.g., a transistor M2). For example, if the transistor **906** (e.g., a transistor M2) is turned off, the charging circuit **910** (e.g., the charging circuit **810**) does not charge the capacitor **668** (e.g., a capacitor C1). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **910** (e.g., the charging circuit **810**) does not charge the capacitor **668** (e.g., a capacitor C1) in order to reduce power consumption of the PWM controller **630** (e.g., an IC chip).

(50) As discussed above and further emphasized here, FIG. 9 is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. For example, the voltage detection circuit **920** is part of the circuit **980** for undervoltage-lockout (UVLO), and the signal **924** is internal to the circuit **980** for undervoltage-lockout (UVLO).

(51) FIG. 10 shows simplified timing diagrams under certain dynamic-load and/or no-load conditions for the power converter **600** that includes the PWM controller **630** as shown in FIG. 6 and FIG. 9 according to some embodiments. These diagrams are merely examples, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The waveform **1030** represents the feedback signal **635** as a function of time, the waveform **1040** represents the drive signal **647** as a function of time, the waveform **1050** represents the voltage **645** (e.g., V.sub.DD) as a function of time, and the waveform **1060** represents the signal **932** as a function of time.

(52) According to some embodiments, the waveforms **1030**, **1040**, **1050**, and **1060** are timing diagrams when the power converter **600** is under the normal operation after the voltage **645** has risen above the startup threshold (e.g., VDD_UVLO). For example, at the beginning of the normal operation, the voltage detection circuit **920** generates the signal **924**, so that the circuit **980** for

undervoltage-lockout (UVLO) generates the signal **921** that indicates the end of the startup process and causes the internal control circuit **990** to power on other components of the PWM controller **630** (e.g., an IC chip). As an example, during the normal operation, other components of the PWM controller **630** (e.g., an IC chip) remains powered on. For example, the higher predetermined threshold (e.g., VDD_HV_OFF) is larger than the lower predetermined threshold (e.g., VDD_HV_ON), and the lower predetermined threshold (e.g., VDD_HV_ON) is larger than the startup threshold (e.g., VDD_UVLO).

(53) As shown in FIG. **10**, at time t.sub.20, the feedback signal **635** becomes smaller than a signal threshold **1034** (e.g., FB_Burst_L), and at time t.sub.25, the feedback signal **635** becomes larger than a signal threshold **1032** (e.g., FB_Burst_H) as shown by the waveform **1030** according to certain embodiments. For example, at time t.sub.26, the feedback signal **635** again becomes smaller than the signal threshold **1034** (e.g., FB_Burst_L) as shown by the waveform **1030**. As an example, the signal threshold **1032** (e.g., FB_Burst_H) is larger than the signal threshold **1034** (e.g., FB_Burst_L) in magnitude.

(54) In some embodiments, before time t.sub.20, the drive signal **647** changes between a logic high level and a logic low level within each clock cycle, and from time t.sub.25 to time t.sub.26, the drive signal **647** changes between the logic high level and the logic low level within each clock cycle as shown by the waveform **1040**. In certain embodiments, from time t.sub.20 to time t.sub.25, the drive signal **647** remains at the logic low level during multiple clock cycles, and after time t.sub.26, the drive signal **647** remains at the logic low level during multiple clock cycles as shown by the waveform **1040**.

(55) As shown by the waveform **1050**, from time t.sub.20 to time t.sub.21, the voltage **645** (e.g., VDD) decreases with time, and at time t.sub.21, the voltage **645** (e.g., V.sub.DD) decreases to the lower predetermined threshold (e.g., VDD_HV_ON) according to some embodiments. For example, at time t.sub.21, in response to the voltage **645** (e.g., V.sub.DD) decreasing to the lower predetermined threshold (e.g., VDD_HV_ON), the signal **932** changes from the logic low level to the logic high level in order to turn on the transistor **906** (e.g., a transistor M2) as shown by the waveform **1060**.

(56) According to certain embodiments, from time t.sub.21 to time t.sub.22, the signal **932** remains at the logic high level as shown by the waveform **1060**, the transistor **906** (e.g., a transistor M2) remains turned on, and the voltage **645** (e.g., V.sub.DD) increases with time as shown by the waveform **1050**. For example, at time t.sub.22, the voltage **645** (e.g., V.sub.DD) increases to the higher predetermined threshold (e.g., VDD_HV_OFF) as shown by the waveform **1050**. As an example, at time t.sub.22, in response to the voltage **645** (e.g., V.sub.DD) increasing to the higher predetermined threshold (e.g., VDD_HV_OFF), the signal **932** changes from the logic high level to the logic low level in order to turn off the transistor **906** (e.g., a transistor M2) as shown by the waveform **1060**.

(57) As shown by the waveform **1050**, from time t.sub.22 to time t.sub.23, the voltage **645** (e.g., VDD) decreases with time, and at time t.sub.23, the voltage **645** (e.g., V.sub.DD) decreases to the lower predetermined threshold (e.g., VDD_HV_ON) according to some embodiments. For example, at time t.sub.23, in response to the voltage **645** (e.g., V.sub.DD) decreasing to the lower predetermined threshold (e.g., VDD_HV_ON), the signal **932** changes from the logic low level to the logic high level in order to turn on the transistor **906** (e.g., a transistor M2) as shown by the waveform **1060**.

(58) According to certain embodiments, from time t.sub.23 to time t.sub.24, the signal **932** remains at the logic high level as shown by the waveform **1060**, the transistor **906** (e.g., a transistor M2) remains turned on, and the voltage **645** (e.g., V.sub.DD) increases with time as shown by the waveform **1050**. For example, at time t.sub.24, the voltage **645** (e.g., V.sub.DD) increases to the higher predetermined threshold (e.g., VDD_HV_OFF) as shown by the waveform **1050**. As an example, at time t.sub.24, in response to the voltage **645** (e.g., V.sub.DD) increasing to the higher

predetermined threshold (e.g., VDD_HV_OFF), the signal **932** changes from the logic high level to the logic low level in order to turn off the transistor **906** (e.g., a transistor M2) as shown by the waveform **1060**.

(59) In some embodiments, as shown by the waveform **1060**, the signal **932** remains at the logic low level before time t.sub.21, change from the logic low level to the logic high level at time t.sub.21, remains at the logic high level from time t.sub.21 to time t.sub.22, changes from the logic high level to the logic low level at time t.sub.22, remains at the logic low level from time t.sub.22 to time t.sub.23, changes from the logic low level to the logic high level at time t.sub.23, remains at the logic high level from time t.sub.23 to time t.sub.24, changes from the logic high level to the logic low level at time t.sub.24, and remains at the logic low level after time t.sub.24.

(60) FIG. **11** shows simplified timing diagrams under certain dynamic-load conditions for the power converter **300** as shown in FIG. **3** according to certain embodiments. The waveform **1110** represents the output current **383** (e.g., I.sub.out) as a function of time, the waveform **1120** represents the output voltage **381** (e.g., V.sub.out) as a function of time, the waveform **1130** represents the feedback signal **337** as a function of time, the waveform **1140** represents the drive signal **345** as a function of time, the waveform **1150** represents the voltage **349** (e.g., V.sub.DD) as a function of time, and the waveform **1160** represents the voltage **335** (e.g., V₁₀) as a function of time.

(61) In some embodiments, before time t.sub.31, the output current **383** (e.g., I.sub.out) is at a current level **1112**, and within each clock cycle, the drive signal **345** changes between a logic high level and a logic low level. For example, at time t.sub.31, the output current **383** (e.g., I.sub.out) decreases from the current level **1112** to a current level **1114**. As an example, from time t.sub.31 to time t.sub.32, the output current **383** (e.g., I.sub.out) remains at the current level **1114**. For example, at time t.sub.32, the output current **383** (e.g., I.sub.out) increases from the current level **1114** to the current level **1112**. As an example, after time t.sub.32, the output current **383** (e.g., I.sub.out) remains at the current level **1112**. In some examples, the current level **1112** is higher than the current level **1114**. For example, if the output current **383** (e.g., I.sub.out) is at the current level **1112**, the power converter **300** is under a high-load condition. As an example, if the output current **383** (e.g., I.sub.out) is at the current level **1114**, the power converter **300** is under a light-load condition. In certain embodiments, from time t.sub.31 to time t.sub.32, the power converter **300** is under the light-load condition. For example, under the light-load condition, after time t.sub.31 but before time t.sub.32, the output voltage **381** (e.g., V.sub.out) overshoots as shown by the waveform **1120**, the feedback signal **337** drops as shown by the waveform **1130**, the drive signal **345** remains at the logic low level during multiple clock cycles as shown by the waveform **1140**, and the voltage **349** (e.g., V.sub.DD) decreases gradually as shown by the waveform **1150**. In certain embodiments, from time t.sub.31 to time t.sub.32, even though the voltage **349** (e.g., V.sub.DD) decreases gradually as shown by the waveform **1150**, the voltage **335** (e.g., V_{sub.10}) remains equal to the reference voltage (e.g., V_{sub.reg}) as shown by the waveform **1160**. For example, the voltage **335** (e.g., V_{sub.10}) is generated by the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) based on the voltage **349** (e.g., V.sub.DD). As an example, the generated voltage **335** (e.g., V_{sub.10}) remains constant as shown by the waveform **1160**.

(62) According to some embodiments, under certain dynamic-load conditions, if the voltage **349** (e.g., V.sub.DD) decreases too much, the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) cannot keep the voltage **335** (e.g., V_{sub.10}) equal to the reference voltage (e.g., V_{sub.reg}) and the generated voltage **335** (e.g., V_{sub.10}) also decreases. For example, if the voltage **335** (e.g., V_{sub.10}) becomes too small in magnitude, the undervoltage protection is triggered for the PWM controller **330** (e.g., an IC chip) because of insufficient power supply.

(63) FIG. **12** shows simplified timing diagrams under no-load condition for the power converter **300** as shown in FIG. **3** according to certain embodiments. The waveform **1230** represents the feedback signal **337** as a function of time, the waveform **1240** represents the drive signal **345** as a

function of time, the waveform **1250** represents the voltage **349** (e.g., VDD) as a function of time, and the waveform **1260** represents the voltage **335** (e.g., V.sub.10) as a function of time.

(64) According to some embodiments, under no-load condition, the output current **383** (e.g., I.sub.out) is equal to zero in magnitude. For example, at time t.sub.40, the feedback signal **337** becomes larger than a signal threshold **1232** (e.g., FB_Burst_H), and at time t.sub.41, the feedback signal **337** becomes smaller than a signal threshold **1234** (e.g., FB_Burst_L) as shown by the waveform **1230**. As an example, at time t.sub.42, the feedback signal **337** becomes larger than the signal threshold **1232** (e.g., FB_Burst_H), and at time t.sub.43, the feedback signal **337** becomes smaller than the signal threshold **1234** (e.g., FB_Burst_L) as shown by the waveform **1230**. For example, the signal threshold **1232** (e.g., FB_Burst_H) is larger than the signal threshold **1234** (e.g., FB_Burst_L) in magnitude.

(65) In some embodiments, from time t.sub.40 to time t.sub.41, the drive signal **345** changes between a logic high level and a logic low level within each clock cycle as shown by the waveform **1240**, and the voltage **349** (e.g., V.sub.DD) increases as shown by the waveform **1250**. In certain embodiments, time t.sub.41 to time t.sub.42, the drive signal **345** remains at the logic low level during multiple clock cycles as shown by the waveform **1240**, and the voltage **349** (e.g., VDD) decreases gradually as shown by the waveform **1250**. In some embodiments, from time t.sub.42 to time t.sub.43, the drive signal **345** changes between the logic high level and the logic low level within each clock cycle as shown by the waveform **1240**, and the voltage **349** (e.g., V.sub.DD) increases as shown by the waveform **1250**.

(66) In certain embodiments, from time t.sub.41 to time t.sub.42, even though the voltage **349** (e.g., V.sub.DD) decreases gradually as shown by the waveform **1250**, the voltage **335** (e.g., V.sub.10) remains equal to the reference voltage (e.g., V.sub.reg) as shown by the waveform **1260**. For example, the voltage **335** (e.g., V.sub.10) is generated by the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) based on the voltage **349** (e.g., V.sub.DD). As an example, the generated voltage **335** (e.g., V.sub.10) remains constant as shown by the waveform **1260**.

(67) According to some embodiments, under no-load condition, if the voltage **349** (e.g., VDD) decreases too much, the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) cannot keep the voltage **335** (e.g., V.sub.10) equal to the reference voltage (e.g., V.sub.reg) and the generated voltage **335** (e.g., V.sub.10) also decreases. For example, if the voltage **335** (e.g., V.sub.10) becomes too small in magnitude, the undervoltage protection is triggered for the PWM controller **330** (e.g., an IC chip) because of insufficient power supply.

(68) FIG. **13** is a simplified diagram showing certain components of the PWM controller **730** of the power converter **700** as shown in FIG. **7** according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The PWM controller **730** includes a charging circuit **1310**, voltage detection circuits **1320** and **1370**, charging controllers **1330** and **1360**, a circuit **1380** for undervoltage-lockout (UVLO), an internal control circuit **1390**, a DC-DC controller **1350**, a transistor **1352**, resistors **1354** and **1356**.

Additionally, the PWM controller **730** (e.g., an IC chip) includes the terminal **732** (e.g., a SW pin), the terminal **734** (e.g., a V10 pin), the terminal **738** (e.g., a GND pin), the terminal **740** (e.g., a HV pin), and the terminal **748** (e.g., a VDD pin). In certain examples, the charging circuit **1310** includes a transistor **1302** (e.g., a junction field-effect transistor), a resistor **1304** (e.g., a resistor R6), a resistor **1378** (e.g., a resistor R7), a transistor **1306** (e.g., a metal-oxide-semiconductor field-effect transistor), a transistor **1376** (e.g., a metal-oxide-semiconductor field-effect transistor), and a diode **1382** (e.g., a Schottky diode). In some examples, the charging circuit **1310** includes two current generators. For example, one current generator of the charging circuit **1310** includes the transistor **1302** (e.g., a junction field-effect transistor), the resistor **1304** (e.g., a resistor R6), and the transistor **1306** (e.g., a metal-oxide-semiconductor field-effect transistor). As an example, another current generator of the charging circuit **1310** includes the transistor **1302** (e.g., a junction

field-effect transistor), the resistor **1378** (e.g., a resistor R7), and the transistor **1376** (e.g., a metal-oxide-semiconductor field-effect transistor). In certain examples, the inductor **716** of the power converter **700** (e.g., an AC-DC power supply) is not part of the PWM controller **730** (e.g., an IC chip). For example, a DC-DC converter includes the inductor **716**, the capacitor **768** (e.g., the capacitor C3), the capacitor **718** (e.g., a capacitor C5), the DC-DC controller **1350**, the transistor **1352** (e.g., a transistor M7), the resistor **1354** (e.g., a resistor R8), the resistor **1356** (e.g., a resistor R9), and the diode **1382**.

(69) According to certain embodiments, a drain terminal of the transistor **1302** (e.g., JFET) is connected to the terminal **740** (e.g., a HV pin), the gate terminal of the transistor **1302** (e.g., JFET) is biased to the ground voltage on the primary side of the power converter **700**, and a source terminal of the transistor **1302** (e.g., JFET) is connected to one terminal of the resistor **1304** (e.g., a resistor R6). For example, another terminal of the resistor **1304** (e.g., a resistor R6) is connected to a source terminal of the transistor **1306** (e.g., a transistor M5). As an example, a gate terminal of the transistor **1306** (e.g., a transistor M5) receives a signal **1332** from the charging controller **1330**. For example, a drain terminal of the transistor **1306** (e.g., a transistor M5) is connected to the terminal **748** (e.g., a VDD pin) and is also connected to one terminal of the inductor **716** through the terminal **748** (e.g., a VDD pin). As an example, the transistor **1306** (e.g., a transistor M5) is a PMOS transistor (e.g., a high-voltage PMOS transistor).

(70) In some examples, another terminal of the inductor **716** is connected to the terminal **732** (e.g., a SW pin) and is also connected to an anode of the diode **1382** (e.g., a diode D1) through the terminal **732** (e.g., a SW pin). For example, a cathode of the diode **1382** (e.g., a diode D1) is connected to the terminal **734** (e.g., a V10 pin), a drain terminal of the transistor **1376** (e.g., a transistor M6), and one terminal of the resistor **1354** (e.g., a resistor R8). As an example, another terminal of the resistor **1354** (e.g., a resistor R8) is connected to an input terminal of the DC-DC controller **1350** and one terminal of the resistor **1356** (e.g., a resistor R9). For example, another terminal of the resistor **1356** (e.g., a resistor R9) is connected to the terminal **738** (e.g., a GND pin), which is biased to the ground voltage on the primary side of the power converter **700**. As an example, the transistor **1376** (e.g., a transistor M6) is a PMOS transistor.

(71) In certain examples, an output terminal of the DC-DC controller **1350** is connected to a gate terminal of the transistor **1352** (e.g., a transistor M7). For example, a source terminal of the transistor **1352** (e.g., a transistor M7) is biased to the ground voltage on the primary side of the power converter **700**. As an example, a drain terminal of the transistor **1352** (e.g., a transistor M7) is connected to the anode of the diode **1382** and the terminal **732** (e.g., a SW pin). For example, the transistor **1352** (e.g., a transistor M7) is an NMOS transistor.

(72) According to some embodiments, the source terminal of the transistor **1302** (e.g., JFET) is connected to one terminal of the resistor **1378** (e.g., a resistor R7). For example, another terminal of the resistor **1378** (e.g., a resistor R7) is connected to a source terminal of the transistor **1376** (e.g., a transistor M6). As an example, the gate terminal of the transistor **1376** (e.g., a transistor M6) is connected to an output terminal of the charging controller **1360**. For example, a drain terminal of the transistor **1376** (e.g., a transistor M6) is connected to the cathode of the diode **1382** (e.g., a diode D1) and the terminal **734** (e.g., a V10 pin). In certain examples, an input terminal of the charging controller **1360** receives a signal **1321** (e.g., Char_v10) from the voltage detection circuit **1320**, and another input terminal of the charging controller **1360** receives a signal **1371** from the voltage detection circuit **1370**.

(73) According to certain embodiments, an input terminal of the voltage detection circuit **1320** is connected to the drain terminal of the transistor **1306** (e.g., a transistor M5) and the terminal **748** (e.g., a VDD pin). For example, the output terminal of the voltage detection circuit **1320** sends the signal **1321** to the charging controller **1360**. In some examples, an input terminal of the voltage detection circuit **1370** is connected to the terminal **734** (e.g., a V10 pin). For example, the voltage detection circuit **1370** generates signals **1322**, **1371**, and **1373**. As an example, the voltage

detection circuit **1370** sends the signal **1322** to the charging controller **1330**, sends the signal **1371** to the charging controller **1360**, and sends the signal **1373** to the circuit **1380** for undervoltage-lockout (UVLO). In certain examples, an output terminal of the circuit **1380** for undervoltage-lockout (UVLO) is connected to an input terminal of the internal control circuit **1390**. As an example, the internal control circuit **1390** includes various control circuits, and the internal control circuit **1390** does not include the charging controller **1330** and does not include the charging controller **1360**. For example, the internal control circuit **1390** includes one or more control circuits for one or more oscillators, one or more control circuits for one or more timers, one or more drivers, one or more logic control circuits, and/or one or more protection circuits.

(74) As shown in FIG. 7, after the EMI filter **722** receives the AC voltage **750**, the PWM controller **730** (e.g., an IC chip) undergoes a startup process according to certain embodiments. In some examples, during the startup process, if the voltage detection circuit **1370** determines that the voltage **735** has not reached a startup threshold (e.g., V_{10_UVLO}), the voltage detection circuit **1370** generates the signal **1322**, so that the charging controller **1330** generates the signal **1332** to turn on the transistor **1306** (e.g., a transistor M5) and the charging circuit **1310** generates a charging current **1392**. For example, the charging current **1392** flows through the transistor **1302** (e.g., JFET), the resistor **1304** (e.g., a resistor R6), and the transistor **1306** (e.g., a transistor M5) to charge the capacitor **768** (e.g., the capacitor C3) and raise the voltage **749** at the terminal **748** (e.g., a VDD pin). As an example, the voltage **749** is used, through the inductor **716** and the diode **1382**, to charge the capacitor **718** (e.g., a capacitor C5) and raise the voltage **735**, until the voltage **735** reaches the startup threshold (e.g., V_{10_UVLO}). In certain examples, before the voltage **735** reaches the startup threshold (e.g., V_{10_UVLO}), the voltage detection circuit **1370** generates the signal **1373**, so that the circuit **1380** for undervoltage-lockout (UVLO) does not generate the signal **1391** that indicates the end of the startup process and causes the internal control circuit **1390** to power on other components of the PWM controller **730** (e.g., an IC chip). For example, during the startup process, other components of the PWM controller **730** (e.g., an IC chip) remain powered off. As an example, during the startup process, the transistor **1376** (e.g., a transistor M6) remains turned off, and the DC-DC converter does not operate.

(75) According to some embodiments, during the startup process, the charging current **1392** flows through the transistor **1302** (e.g., JFET), the resistor **1304** (e.g., a resistor R6), and the transistor **1306** (e.g., a transistor M5) to raise the voltage **749**, and the voltage **749** is used, through the inductor **716** and the diode **1382**, to raise the voltage **735**. As an example, the charging current **1392** is determined as follows:

$$(76) \quad I_{hv_vdd} = \frac{V_{HVS} - V_{DD}}{R_6} \quad (\text{Equation 3})$$

where $I_{sub.hv_vdd}$ represents the charging current **1392**. Additionally, $V_{sub.HVS}$ represents the voltage **741**, and $V_{sub.DD}$ represents the voltage **749**. Also, $R_{sub.6}$ represents the resistance of the resistor **1304**. For example, according to Equation 3, the charging current **1392** (e.g., $I_{sub.hv_vdd}$) decreases with the increasing voltage **749** (e.g., $V_{sub.DD}$).

(77) In certain embodiments, if the voltage detection circuit **1370** determines that the voltage **735** has risen above the startup threshold (e.g., V_{10_UVLO}), the voltage detection circuit **1370** generates the signal **1322**, so that the charging controller **1330** generates the signal **1332** to turn off the transistor **1306** (e.g., a transistor M5) and the charging circuit **1310** does not charge the capacitor **718** (e.g., a capacitor C5). In some examples, after the voltage detection circuit **1370** determines that the voltage **735** has risen above the startup threshold (e.g., V_{10_UVLO}), the startup process of the PWM controller **730** (e.g., an IC chip) ends and the normal operation for the power converter **700** starts. For example, at the beginning of the normal operation, the voltage detection circuit **1370** generates the signal **1373**, so that the circuit **1380** for undervoltage-lockout (UVLO) generates the signal **1391** that indicates the end of the startup process and causes the internal control circuit **1390** to power on other components of the PWM controller **730** (e.g., an IC chip).

As an example, during the normal operation, other components of the PWM controller **730** (e.g., an IC chip) remain powered on. In certain examples, under normal operation, the transistor **1306** (e.g., a transistor **M5**) remains turned off. In some examples, the voltage detection circuit **1370** determines whether the voltage **735** has risen to the startup threshold (e.g., **V10_UVLO**) during the startup process, and also determines whether the voltage **735** rises to a predetermined threshold (e.g., **V10_HV_OFF**) during the normal operation.

(78) In some embodiments, during the normal operation, when the voltage **749** (e.g., **VDD**) is higher than a reference voltage (e.g., **V.sub.reg**), the DC-DC converter is idle, and the auxiliary winding **706** charges the capacitor **768** (e.g., a capacitor **C3**) to support the voltage **749** (e.g., **V.sub.DD**), which is used to charge the capacitor **718** (e.g., a capacitor **C5**) and support the voltage **735** so that the voltage **735** at the terminal **734** (e.g., a **V10** pin) is equal to the voltage **749** (e.g., **V.sub.DD**) minus the forward bias voltage of the diode **1382**. In some examples, during the normal operation, when the voltage **749** (e.g., **V.sub.DD**) is smaller than the reference voltage (e.g., **V.sub.reg**) but is not smaller than a lower predetermined threshold (e.g., **VDD_HV_ON**), the voltage **735** at the terminal **734** (e.g., a **V10** pin) is generated by the DC-DC converter using the voltage **749** (e.g., **V.sub.DD**), and the generated voltage **735** is larger than the voltage **749** (e.g., **V.sub.DD**) and is equal to the reference voltage (e.g., **V.sub.reg**).

(79) According to certain embodiments, during the normal operation, if the voltage detection circuit **1320** determines that the voltage **749** drops to the lower predetermined threshold (e.g., **VDD_HV_ON**), the voltage detection circuit **1320** generates the signal **1321** (e.g., **Char_v10**), so that the charging controller **1360** generates a signal **1361** (e.g., **Char_en**) to turn on the transistor **1376** (e.g., a transistor **M6**). In some examples, if the transistor **1376** (e.g., a transistor **M6**) is turned on, the charging circuit **1310** generates a charging current **1394** to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735**. For example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1310** generates the charging current **1394** to flow through the terminal **734** (e.g., a **V10** pin) and to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735**, in order to prevent insufficient power supply to the PWM controller **730** (e.g., an IC chip). In certain examples, the charging current **1394** is determined as follows:

$$(80) \quad I_{hv_v10} = \frac{V_{HVS} - V_{10}}{R_7} \quad (\text{Equation4})$$

where I_{hv_v10} represents the charging current **1394**. Additionally, **V.sub.HVS** represents the voltage **741**, and **V.sub.10** represents the voltage **735**. Also, **R.sub.7** represents the resistance of the resistor **1378**.

(81) In some embodiments, during the normal operation, if the voltage detection circuit **1320** determines that the voltage **749** rises to a higher predetermined threshold (e.g., **VDD_HV_OFF**), the voltage detection circuit **1320** generates the signal **1321**, so that the charging controller **1360** generates the signal **1361** (e.g., **Char_en**) to turn off the transistor **1376** (e.g., a transistor **M6**). For example, if the transistor **1376** (e.g., a transistor **M6**) is turned off, the charging circuit **1310** does not charge the capacitor **718** (e.g., a capacitor **C5**). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1310** does not charge the capacitor **718** (e.g., a capacitor **C5**) in order to reduce power consumption of the PWM controller **730** (e.g., an IC chip).

(82) In certain embodiments, during the normal operation, the voltage detection circuit **1370** determines whether the voltage **735** rises to the predetermined threshold (e.g., **V10_HV_OFF**) and generates the signal **1371**. For example, if the voltage **735** has not risen to the predetermined threshold (e.g., **V10_HV_OFF**), the signal **1371** is at a first logic level (e.g., a logic low level). As an example, if the voltage **735** has risen to the predetermined threshold (e.g., **V10_HV_OFF**), the signal **1371** is at a second logic level (e.g., a logic high level). In certain examples, if the signal **1371** indicates that the voltage **735** has risen to the predetermined threshold (e.g., **V10_HV_OFF**),

the charging controller **1360** generates the signal **1361** (e.g., Char_en) to turn off the transistor **1376** (e.g., a transistor M6). For example, if the transistor **1376** (e.g., a transistor M6) is turned off, the charging circuit **1310** does not charge the capacitor **718** (e.g., a capacitor C5). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1310** does not charge the capacitor **718** (e.g., a capacitor C5) in order to reduce power consumption of the PWM controller **730** (e.g., an IC chip).

(83) As shown in FIG. **13**, the charging controller **1360** receives the signal **1321** and the signal **1371** and generates the signal **1361** (e.g., Char_en) to turn on or turn off the transistor **1376** (e.g., a transistor M6) according to some embodiments. For example, if the signal **1321** indicates that the voltage **749** drops to the lower predetermined threshold (e.g., VDD_HV_ON), the charging controller **1360** generates the signal **1361** (e.g., Char_en) to turn on the transistor **1376** (e.g., a transistor M6) for generating the charging current **1394** to charge the capacitor **718** (e.g., a capacitor C5) and raise the voltage **735**. As an example, if the signal **1321** indicates that the voltage **749** rises to the higher predetermined threshold (e.g., VDD_HV_OFF), the charging controller **1360** generates the signal **1361** (e.g., Char_en) to turn off the transistor **1376** (e.g., a transistor M6). For example, if the signal **1371** indicates that the voltage **735** has risen to the predetermined threshold (e.g., V10_HV_OFF), the charging controller **1360** generates the signal **1361** (e.g., Char_en) to turn off the transistor **1376** (e.g., a transistor M6).

(84) As discussed above and further emphasized here, FIG. **13** is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. In some examples, the PWM controller **730** also includes a reference-current and reference-voltage generator that receives a signal from the voltage detection circuit **1370** and is connected to the internal control circuit **1390**. In certain examples, the voltage detection circuit **1370** outputs a signal for overvoltage protection. In certain examples, the diode **1382** (e.g., a Schottky diode) is replaced by a parasitic body diode of a PMOS transistor and/or by a PMOS switch. In some examples, the PWM controller **730** also includes a timer that, after being triggered, during a predetermined duration, prevents the signal **1371** from causing the charging controller **1360** to turn off the transistor **1376** (e.g., a transistor M6).

(85) FIG. **14** shows simplified timing diagrams under certain dynamic-load and/or no-load conditions for the power converter **700** that includes the PWM controller **730** as shown in FIG. **7** and FIG. **13** according to some embodiments of the present invention. These diagrams are merely examples, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The waveform **1430** represents the feedback signal **737** as a function of time, the waveform **1440** represents the drive signal **745** as a function of time, the waveform **1450** represents the voltage **749** (e.g., V.sub.DD) as a function of time, the waveform **1460** represents the signal **1361** (e.g., Char_en) as a function of time, and the waveform **1470** represents the voltage **735** (e.g., V10) as a function of time.

(86) According to some embodiments, the waveforms **1430**, **1440**, **1450**, **1460** and **1470** are timing diagrams when the power converter **700** is under the normal operation after the voltage **735** has risen above the startup threshold (e.g., V10_UVLO). For example, at the beginning of the normal operation, the voltage detection circuit **1370** generates the signal **1373**, so that the circuit **1380** for undervoltage-lockout (UVLO) generates the signal **1391** that indicates the end of the startup process and causes the internal control circuit **1390** to power on other components of the PWM controller **730** (e.g., an IC chip). As an example, during the normal operation, other components of the PWM controller **730** (e.g., an IC chip) remains powered on.

(87) As shown in FIG. **10**, at time t.sub.51, the voltage **749** (e.g., V.sub.DD) drops to the lower predetermined threshold (e.g., VDD_HV_ON) as shown by the waveform **1450**, and in response, the transistor **1306** (e.g., the transistor M8) is turned on in order to charge the capacitor **718** (e.g., a capacitor C5) and raise the voltage **735** (e.g., V.sub.10), according to certain embodiments. In some examples, from time t.sub.51 to time t.sub.52, the voltage **735** (e.g., V.sub.10) increases with time

as shown by the waveform **1470**. In certain examples, at time $t_{\text{sub.52}}$, the voltage **735** (e.g., $V_{\text{sub.10}}$) rises to the predetermined threshold (e.g., $V_{10_HV_OFF}$) and in response, the transistor **1306** (e.g., the transistor **M8**) is turned off.

(88) According to some embodiments, at time $t_{\text{sub.61}}$, the voltage **749** (e.g., $V_{\text{sub.DD}}$) drops to the lower predetermined threshold (e.g., $V_{DD_HV_ON}$) as shown by the waveform **1450**, and in response, the transistor **1306** (e.g., the transistor **M8**) is turned on in order to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735** (e.g., $V_{\text{sub.10}}$). In certain examples, from time $t_{\text{sub.61}}$ to time $t_{\text{sub.62}}$, the voltage **735** (e.g., $V_{\text{sub.10}}$) increases with time as shown by the waveform **1470**. In some examples, at time $t_{\text{sub.62}}$, the voltage **749** (e.g., $V_{\text{sub.DD}}$) rises to the higher predetermined threshold (e.g., $V_{DD_HV_OFF}$) as shown by the waveform **1450** and in response, the transistor **1306** (e.g., the transistor **M8**) is turned off.

(89) FIG. **15** is a simplified diagram showing certain components of the PWM controller **730** of the power converter **700** as shown in FIG. **7** according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The PWM controller **730** includes a charging circuit **1510**, voltage detection circuits **1520** and **1570**, charging controllers **1530** and **1560**, a circuit **1580** for undervoltage-lockout (UVLO), an internal control circuit **1590**, a DC-DC controller **1550**, a transistor **1552**, resistors **1554** and **1556**.

Additionally, the PWM controller **730** (e.g., an IC chip) includes the terminal **732** (e.g., a SW pin), the terminal **734** (e.g., a V_{10} pin), the terminal **738** (e.g., a GND pin), the terminal **740** (e.g., a HV pin), and the terminal **748** (e.g., a VDD pin). In certain examples, the charging circuit **1510** includes a transistor **1502** (e.g., a junction field-effect transistor), a resistor **1578** (e.g., a resistor **R7**), a transistor **1506** (e.g., a metal-oxide-semiconductor field-effect transistor), a transistor **1576** (e.g., a metal-oxide-semiconductor field-effect transistor), a diode **1582** (e.g., a Schottky diode), a resistor **2510** (e.g., a resistor **R1**), a resistor **2520** (e.g., a resistor **R2**), a resistor **2530** (e.g., a resistor **R3**), a switch **2512** (e.g., a switch **S1**), a switch **2522** (e.g., a switch **S2**), and a switch **2532** (e.g., a switch **S3**). In some examples, the charging circuit **1510** includes two current generators. For example, one current generator of the charging circuit **1510** includes the transistor **1502** (e.g., a junction field-effect transistor), the resistor **2510** (e.g., a resistor **R1**), the resistor **2520** (e.g., a resistor **R2**), the resistor **2530** (e.g., a resistor **R3**), the switch **2512** (e.g., a switch **S1**), the switch **2522** (e.g., a switch **S2**), the switch **2532** (e.g., a switch **S3**), and the transistor **1506** (e.g., a metal-oxide-semiconductor field-effect transistor). As an example, another current generator of the charging circuit **1510** includes the transistor **1502** (e.g., a junction field-effect transistor), the resistor **1578** (e.g., a resistor **R7**), and the transistor **1576** (e.g., a metal-oxide-semiconductor field-effect transistor). In certain examples, the inductor **716** of the power converter **700** (e.g., an AC-DC power supply) is not part of the PWM controller **730** (e.g., an IC chip). For example, a DC-DC converter includes the inductor **716**, the capacitor **768** (e.g., the capacitor **C3**), the capacitor **718** (e.g., a capacitor **C5**), the DC-DC controller **1550**, the transistor **1552** (e.g., a transistor **M7**), the resistor **1554** (e.g., a resistor **R8**), the resistor **1556** (e.g., a resistor **R9**), and the diode **1582**.

(90) According to certain embodiments, a drain terminal of the transistor **1502** (e.g., JFET) is connected to the terminal **740** (e.g., a HV pin), the gate terminal of the transistor **1502** (e.g., JFET) is biased to the ground voltage on the primary side of the power converter **700**, and a source terminal of the transistor **1502** (e.g., JFET) is connected to one terminal of the resistor **2510** (e.g., a resistor **R1**), one terminal of the resistor **2520** (e.g., a resistor **R2**), and one terminal of the resistor **2530** (e.g., a resistor **R3**). For example, another terminal of the resistor **2510** (e.g., a resistor **R1**) is connected to one terminal of the switch **2512** (e.g., a switch **S1**), another terminal of the resistor **2520** (e.g., a resistor **R2**) is connected to one terminal of the switch **2522** (e.g., a switch **S2**), and another terminal of the resistor **2530** (e.g., a resistor **R3**) is connected to one terminal of the switch **2532** (e.g., a switch **S3**). As an example, another terminal of the switch **2512** (e.g., a switch **S1**), another terminal of the switch **2522** (e.g., a switch **S2**), and another terminal of the switch **2532**

(e.g., a switch S3) are all connected to a source terminal of the transistor **1506** (e.g., a transistor M5). For example, the transistor **1506** (e.g., a transistor M5) is a PMOS transistor (e.g., a high-voltage PMOS transistor).

(91) In some examples, the switch **2512** (e.g., a switch S1) receives a signal **2514** that is used to close (e.g., turn on) or open (e.g., turn off) the switch **2512** (e.g., a switch S1), the switch **2522** (e.g., a switch S2) receives a signal **2524** that is used to close (e.g., turn on) or open (e.g., turn off) the switch **2522** (e.g., a switch S1), and the switch **2532** (e.g., a switch S3) receives a signal **2534** that is used to close (e.g., turn on) or open (e.g., turn off) the switch **2532** (e.g., a switch S1). In certain examples, a gate terminal of the transistor **1506** (e.g., a transistor M5) receives a signal **1532** from the charging controller **1530**. For example, a drain terminal of the transistor **1506** (e.g., a transistor M5) is connected to the terminal **748** (e.g., a VDD pin) and is also connected to one terminal of the inductor **716** through the terminal **748** (e.g., a VDD pin).

(92) In some embodiments, another terminal of the inductor **716** is connected to the terminal **732** (e.g., a SW pin) and is also connected to an anode of the diode **1582** (e.g., a diode D1) through the terminal **732** (e.g., a SW pin). For example, a cathode of the diode **1582** (e.g., a diode D1) is connected to the terminal **734** (e.g., a V10 pin), a drain terminal of the transistor **1576** (e.g., a transistor M6), and one terminal of the resistor **1554** (e.g., a resistor R8). As an example, another terminal of the resistor **1554** (e.g., a resistor R8) is connected to an input terminal of the DC-DC controller **1550** and one terminal of the resistor **1556** (e.g., a resistor R9). For example, another terminal of the resistor **1556** (e.g., a resistor R9) is connected to the terminal **738** (e.g., a GND pin), which is biased to the ground voltage on the primary side of the power converter **700**. As an example, the transistor **1576** (e.g., a transistor M6) is a PMOS transistor.

(93) In certain examples, an output terminal of the DC-DC controller **1550** is connected to a gate terminal of the transistor **1552** (e.g., a transistor M7). For example, a source terminal of the transistor **1552** (e.g., a transistor M7) is biased to the ground voltage on the primary side of the power converter **700**. As an example, a drain terminal of the transistor **1552** (e.g., a transistor M7) is connected to the anode of the diode **1582** and the terminal **732** (e.g., a SW pin). For example, the transistor **1552** (e.g., a transistor M7) is an NMOS transistor.

(94) According to certain embodiments, the source terminal of the transistor **1502** (e.g., JFET) is connected to one terminal of the resistor **1578** (e.g., a resistor R7). For example, another terminal of the resistor **1578** (e.g., a resistor R7) is connected to a source terminal of the transistor **1576** (e.g., a transistor M6). As an example, a gate terminal of the transistor **1576** (e.g., a transistor M6) is connected to an output terminal of the charging controller **1560**. For example, a drain terminal of the transistor **1576** (e.g., a transistor M6) is connected to the cathode of the diode **1582** (e.g., a diode D1) and the terminal **734** (e.g., a V10 pin). In certain examples, an input terminal of the charging controller **1560** receives a signal **1521** (e.g., Char_v10) from the voltage detection circuit **1520**, and another input terminal of the charging controller **1560** receives a signal **1571** from the voltage detection circuit **1570**.

(95) According to some embodiments, an input terminal of the voltage detection circuit **1520** is connected to the drain terminal of the transistor **1506** (e.g., a transistor M5) and the terminal **748** (e.g., a VDD pin). For example, the output terminal of the voltage detection circuit **1520** sends the signal **1521** to the charging controller **1560**. In some examples, an input terminal of the voltage detection circuit **1570** is connected to the terminal **734** (e.g., a V10 pin). For example, the voltage detection circuit **1570** generates signals **1522**, **1571**, and **1573**. As an example, the voltage detection circuit **1570** sends the signal **1522** to the charging controller **1530**, sends the signal **1571** to the charging controller **1560**, and sends the signal **1573** to the circuit **1580** for undervoltage-lockout (UVLO). In certain examples, an output terminal of the circuit **1580** for undervoltage-lockout (UVLO) is connected to an input terminal of the internal control circuit **1590**. As an example, the internal control circuit **1590** includes various control circuits, and the internal control circuit **1590** does not include the charging controller **1530** and does not include the charging

controller **1560**. For example, the internal control circuit **1590** includes one or more control circuits for one or more oscillators, one or more control circuits for one or more timers, one or more drivers, one or more logic control circuits, and/or one or more protection circuits.

(96) As shown in FIG. 7, after the EMI filter **722** receives the AC voltage **750**, the PWM controller **730** (e.g., an IC chip) undergoes a startup process according to certain embodiments. In some examples, during the startup process, if the voltage detection circuit **1570** determines that the voltage **735** has not reached a startup threshold (e.g., V_{10_UVLO}), the voltage detection circuit **1570** generates the signal **1522**, so that the charging controller **1530** generates the signal **1532** to turn on the transistor **1506** (e.g., a transistor M5). For example, if the voltage detection circuit **1520** determines that the voltage **749** is smaller than a first reference voltage (e.g., V_1), the voltage detection circuit **1520** generates the signal **2514** to close (e.g., turn on) the switch **2512**, generates the signal **2524** to open (e.g., turn off) the switch **2522**, and generates the signal **2534** to open (e.g., turn off) the switch **2532**. As an example, if the voltage detection circuit **1520** determines that the voltage **749** is larger than the first reference voltage (e.g., V_1) but smaller than a second reference voltage (e.g., V_2), the voltage detection circuit **1520** generates the signal **2514** to close (e.g., turn on) the switch **2512**, generates the signal **2524** to close (e.g., turn on) the switch **2522**, and generates the signal **2534** to open (e.g., turn off) the switch **2532**, where the second reference voltage (e.g., V_2) is larger than the first reference voltage (e.g., V_1). For example, if the voltage detection circuit **1520** determines that the voltage **749** is larger than both the first reference voltage (e.g., V_1) and the second reference voltage (e.g., V_2), the voltage detection circuit **1520** generates the signal **2514** to close (e.g., turn on) the switch **2512**, generates the signal **2524** to close (e.g., turn on) the switch **2522**, and generates the signal **2534** to close (e.g., turn on) the switch **2532**.

(97) In certain examples, during the startup process, if the voltage **735** has not reached the startup threshold (e.g., V_{10_UVLO}), the charging circuit **1510** generates a charging current **1592** to charge the capacitor **768** (e.g., the capacitor C3) and raise the voltage **749** at the terminal **748** (e.g., a VDD pin). For example, the charging current **1592** is determined as follows:

$$(98) \quad I_{hv_vdd} = \begin{cases} \frac{V_{HVS} - V_{DD}}{R_1} & V_{DD} < V_1 \\ \frac{V_{HVS} - V_{DD}}{R_1 // R_2} & V_1 < V_{DD} < V_2 \\ \frac{V_{HVS} - V_{DD}}{R_1 // R_2 // R_3} & V_{DD} > V_2 \end{cases} \quad (\text{Equation 5})$$

where I_{hv_vdd} represents the charging current **1592**. Additionally, V_{HVS} represents the voltage **741**, and V_{DD} represents the voltage **749**. Moreover, $R_{sub.1}$ represents the resistance of the resistor **2510**, $R_{sub.2}$ represents the resistance of the resistor **2520**, and $R_{sub.3}$ represents the resistance of the resistor **2530**. Also, $R_{sub.1}/R_{sub.2}$ represents the resistance of a combination of the resistors **2510** and **2520**, which are connected in parallel, and $R_{sub.1}/R_{sub.2}/R_{sub.3}$ represents the resistance of a combination of the resistors **2510**, **2520** and **2530**, which are connected in parallel. Additionally, $V_{sub.1}$ represents the first reference voltage, and $V_{sub.2}$ represents the second reference voltage.

(99) In some examples, if the voltage **735** is smaller than the startup threshold (e.g., V_{10_UVLO}) and the voltage **749** is smaller than the first reference voltage (e.g., V_1), the charging current **1592** flows through the transistor **1502** (e.g., FET), the resistor **2510** (e.g., a resistor R1), and the transistor **1506** (e.g., a transistor M5). For example, if the voltage **735** is smaller than the startup threshold (e.g., V_{10_UVLO}) and the voltage **749** is larger than the first reference voltage (e.g., V_1) but smaller than the second reference voltage (e.g., V_2), the charging current **1592** flows through the transistor **1502** (e.g., FET), the resistor **2510** (e.g., a resistor R1), the resistor **2520** (e.g., a resistor R2), and the transistor **1506** (e.g., a transistor M5). As an example, if the voltage **735** is smaller than the startup threshold (e.g., V_{10_UVLO}) and the voltage **749** is larger than both the first reference voltage (e.g., V_1) and the second reference voltage (e.g., V_2), the charging current **1592** flows through the transistor **1502** (e.g., FET), the resistor **2510** (e.g., a resistor R1), the

resistor **2520** (e.g., a resistor R2), the resistor **2530** (e.g., a resistor R3), and the transistor **1506** (e.g., a transistor M5).

(100) According to some embodiments, before the voltage **735** reaches the startup threshold (e.g., **V10_UVLO**), the voltage detection circuit **1570** generates the signal **1573**, so that the circuit **1580** for undervoltage-lockout (UVLO) does not generate the signal **1591** that indicates the end of the startup process and causes the internal control circuit **1590** to power on other components of the PWM controller **730** (e.g., an IC chip). For example, during the startup process, other components of the PWM controller **730** (e.g., an IC chip) remain powered off. As an example, during the startup process, the transistor **1576** (e.g., a transistor M6) remains turned off, and the DC-DC converter does not operate.

(101) In certain embodiments, if the voltage detection circuit **1570** determines that the voltage **735** has risen above the startup threshold (e.g., **V10_UVLO**), the voltage detection circuit **1570** generates the signal **1522**, so that the charging controller **1530** generates the signal **1532** to turn off the transistor **1506** (e.g., a transistor M5) and the charging circuit **1510** does not charge the capacitor **718** (e.g., a capacitor C5). In some examples, after the voltage detection circuit **1570** determines that the voltage **735** has risen above the startup threshold (e.g., **V10_UVLO**), the startup process of the PWM controller **730** (e.g., an IC chip) ends and the normal operation for the power converter **700** starts. For example, at the beginning of the normal operation, the voltage detection circuit **1570** generates the signal **1573**, so that the circuit **1580** for undervoltage-lockout (UVLO) generates the signal **1591** that indicates the end of the startup process and causes the internal control circuit **1590** to power on other components of the PWM controller **730** (e.g., an IC chip). As an example, during the normal operation, other components of the PWM controller **730** (e.g., an IC chip) remain powered on. In certain examples, under normal operation, the transistor **1506** (e.g., a transistor M5) remains turned off. In some examples, the voltage detection circuit **1570** determines whether the voltage **735** has risen to the startup threshold (e.g., **V10_UVLO**) during the startup process, and also determines whether the voltage **735** rises to a predetermined threshold (e.g., **V10_HV_OFF**) during the normal operation.

(102) According to certain embodiments, during the normal operation, when the voltage **749** (e.g., **V.sub.DD**) is higher than a reference voltage (e.g., **V.sub.reg**), the DC-DC converter is idle, and the auxiliary winding **706** charges the capacitor **768** (e.g., a capacitor C3) to support the voltage **749** (e.g., **V.sub.DD**), which is used to charge the capacitor **718** (e.g., a capacitor C5) and support the voltage **735** so that the voltage **735** at the terminal **734** (e.g., a **V10** pin) is equal to the voltage **749** (e.g., **V.sub.DD**) minus the forward bias voltage of the diode **1582**. In some examples, during the normal operation, when the voltage **749** (e.g., **V.sub.DD**) is smaller than the reference voltage (e.g., **V.sub.reg**) but is not smaller than a lower predetermined threshold (e.g., **VDD_HV_ON**), the voltage **735** at the terminal **734** (e.g., a **V10** pin) is generated by the DC-DC converter using the voltage **749** (e.g., **V.sub.DD**), and the generated voltage **735** is larger than the voltage **749** (e.g., **V.sub.DD**) and is equal to the reference voltage (e.g., **V.sub.reg**).

(103) According to some embodiments, during the normal operation, if the voltage detection circuit **1520** determines that the voltage **749** drops to the lower predetermined threshold (e.g., **VDD_HV_ON**), the voltage detection circuit **1520** generates the signal **1521** (e.g., **Char_v10**), so that the charging controller **1560** generates a signal **1561** (e.g., **Char_en**) to turn on the transistor **1576** (e.g., a transistor M6). In some examples, if the transistor **1576** (e.g., a transistor M6) is turned on, the charging circuit **1510** generates a charging current **1594** to charge the capacitor **718** (e.g., a capacitor C5) and raise the voltage **735**. For example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1510** generates the charging current **1594** to flow through the terminal **734** (e.g., a **V10** pin) and to charge the capacitor **718** (e.g., a capacitor C5) and raise the voltage **735**, in order to prevent insufficient power supply to the PWM controller **730** (e.g., an IC chip). In certain examples, the charging current **1594** is determined as follows:

$$(104) \ I_{\text{hv_v10}} = \frac{V_{\text{HVS}} - V_{10}}{R_7} \quad (\text{Equation6})$$

where $I_{\text{sub.hv_v10}}$ represents the charging current **1594**. Additionally, $V_{\text{sub.HVS}}$ represents the voltage **741**, and $V_{\text{sub.10}}$ represents the voltage **735**. Also, $R_{\text{sub.7}}$ represents the resistance of the resistor **1578**.

(105) In some embodiments, during the normal operation, if the voltage detection circuit **1520** determines that the voltage **749** rises to a higher predetermined threshold (e.g., $V_{\text{DD_HV_OFF}}$), the voltage detection circuit **1520** generates the signal **1521**, so that the charging controller **1560** generates the signal **1561** (e.g., Char_en) to turn off the transistor **1576** (e.g., a transistor **M6**). For example, if the transistor **1576** (e.g., a transistor **M6**) is turned off, the charging circuit **1510** does not charge the capacitor **718** (e.g., a capacitor **C5**). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1510** does not charge the capacitor **718** (e.g., a capacitor **C5**) in order to reduce power consumption of the PWM controller **730** (e.g., an IC chip).

(106) In certain embodiments, during the normal operation, the voltage detection circuit **1570** determines whether the voltage **735** rises to the predetermined threshold (e.g., $V_{10_HV_OFF}$) and generates the signal **1571**. For example, if the voltage **735** has not risen to the predetermined threshold (e.g., $V_{10_HV_OFF}$), the signal **1571** is at a first logic level (e.g., a logic low level). As an example, if the voltage **735** has risen to the predetermined threshold (e.g., $V_{10_HV_OFF}$), the signal **1571** is at a second logic level (e.g., a logic high level). In certain examples, if the signal **1571** (e.g., Char_en) indicates that the voltage **735** has risen to the predetermined threshold (e.g., $V_{10_HV_OFF}$), the charging controller **1560** generates the signal **1561** (e.g., Char_en) to turn off the transistor **1576** (e.g., a transistor **M6**). For example, if the transistor **1576** (e.g., a transistor **M6**) is turned off, the charging circuit **1510** does not charge the capacitor **718** (e.g., a capacitor **C5**). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1510** does not charge the capacitor **718** (e.g., a capacitor **C5**) in order to reduce power consumption of the PWM controller **730** (e.g., an IC chip).

(107) As shown in FIG. **15**, the charging controller **1560** receives the signals **1521** and **1571** and generates the signal **1561** (e.g., Char_en) to turn on or turn off the transistor **1576** (e.g., a transistor **M6**) according to some embodiments. For example, if the signal **1521** indicates that the voltage **749** drops to the lower predetermined threshold (e.g., $V_{\text{DD_HV_ON}}$), the charging controller **1560** generates the signal **1561** (e.g., Char_en) to turn on the transistor **1576** (e.g., a transistor **M6**) for generating the charging current **1594** to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735**. As an example, if the signal **1521** indicates that the voltage **749** rises to the higher predetermined threshold (e.g., $V_{\text{DD_HV_OFF}}$), the charging controller **1560** generates the signal **1561** (e.g., Char_en) to turn off the transistor **1576** (e.g., a transistor **M6**). For example, if the signal **1571** indicates that the voltage **735** has risen to the predetermined threshold (e.g., $V_{10_HV_OFF}$), the charging controller **1560** generates the signal **1561** (e.g., Char_en) to turn off the transistor **1576** (e.g., a transistor **M6**).

(108) As discussed above and further emphasized here, FIG. **15** is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. In some examples, the PWM controller **730** also includes a reference-current and reference-voltage generator that receives a signal from the voltage detection circuit **1570** and is connected to the internal control circuit **1590**. In certain examples, the voltage detection circuit **1570** outputs a signal for overvoltage protection. In certain examples, the diode **1582** (e.g., a Schottky diode) is replaced by a parasitic body diode of a PMOS transistor and/or by a PMOS switch. In some examples, the PWM controller **730** also includes a timer that, after being triggered, during a predetermined duration, prevents the signal **1571** from causing the charging controller **1560** to turn off the transistor **1576** (e.g., a transistor **M6**).

(109) FIG. **16** shows a simplified diagram showing the charging current **1592** as a function of the

voltage **749** for the power converter **700** that includes the PWM controller **730** as shown in FIG. 7 and FIG. 15 according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown in FIG. 16, I.sub.hv_vdd represents the charging current **1592**, and V.sub.DD represents the voltage **749** according to certain embodiments. In some examples, V.sub.1 represents the first reference voltage, and V.sub.2 represents the second reference voltage. In certain examples, V.sub.c represents a voltage magnitude of the voltage **749**, at which the startup process of the PWM controller **730** (e.g., an IC chip) ends. For example, V.sub.c represents the voltage magnitude of the voltage **749**, at which the voltage detection circuit **1570** determines that the voltage **735** has risen above the startup threshold (e.g., V₁₀_UVLO). As an example, V.sub.HVS represents the voltage **741**, R.sub.1 represents the resistance of the resistor **2510**, R.sub.2 represents the resistance of the resistor **2520**, and R.sub.3 represents the resistance of the resistor **2530**.

(110) In certain embodiments, according to Equation 5, if V.sub.DD is equal to zero, I.sub.hv_vdd is equal to

$$(111) \frac{V_{HVS}}{R_1},$$

and when V.sub.DD increases from zero to V.sub.1, I.sub.hv_vdd decreases with the increasing V.sub.DD. In some embodiments, according to Equation 5, if V.sub.DD reaches V.sub.1, I.sub.hv_vdd rises to

$$(112) \frac{V_{HVS} - V_1}{R_1 // R_2},$$

and when V.sub.DD increases from V.sub.1 to V.sub.2, I.sub.hv_vdd decreases with the increasing V.sub.DD. In certain embodiments, according to Equation 5, if V.sub.DD reaches V.sub.2,

I.sub.hv_vdd rises to

$$(113) \frac{V_{HVS} - V_2}{R_1 // R_2 // R_3},$$

and when V.sub.DD increases from V.sub.2 to V.sub.c, I.sub.hv_vdd decreases with the increasing V.sub.DD. In some embodiments, if V.sub.DD reaches V.sub.c, the startup process of the PWM controller **730** (e.g., an IC chip) ends and I.sub.hv_vdd drops to zero.

(114) FIG. 17 is a simplified diagram showing certain components of the PWM controller **730** of the power converter **700** as shown in FIG. 7 according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The PWM controller **730** includes a charging circuit **1710**, voltage detection circuits **1720** and **1770**, a charging controller **1730**, a circuit **1780** for undervoltage-lockout (UVLO), an internal control circuit **1790**, a DC-DC controller **1750**, a transistor **1752**, resistors **1754** and **1756**. Additionally, the PWM controller **730** (e.g., an IC chip) includes the terminal **732** (e.g., a SW pin), the terminal **734** (e.g., a V₁₀ pin), the terminal **738** (e.g., a GND pin), the terminal **740** (e.g., a HV pin), and the terminal **748** (e.g., a VDD pin). In certain examples, the charging circuit **1710** includes a transistor **1702** (e.g., a junction field-effect transistor), a transistor **1706** (e.g., a metal-oxide-semiconductor field-effect transistor), a diode **1782** (e.g., a Schottky diode), a resistor **2710** (e.g., a resistor R₁), a resistor **2720** (e.g., a resistor R₂), a resistor **2730** (e.g., a resistor R₃), a switch **2712** (e.g., a switch S₁), a switch **2722** (e.g., a switch S₂), and a switch **2732** (e.g., a switch S₃). As an example, the inductor **716** of the power converter **700** (e.g., an AC-DC power supply) is not part of the PWM controller **730** (e.g., an IC chip). For example, a DC-DC converter includes the inductor **716**, the capacitor **768** (e.g., the capacitor C₃), the capacitor **718** (e.g., a capacitor C₅), the DC-DC controller **1750**, the transistor **1752** (e.g., a transistor M₄), the resistor **1754** (e.g., a resistor R₄), the resistor **1756** (e.g., a resistor R₅), and the diode **1782**.

(115) According to certain embodiments, a drain terminal of the transistor **1702** (e.g., JFET) is connected to the terminal **740** (e.g., a HV pin), the gate terminal of the transistor **1702** (e.g., JFET) is biased to the ground voltage on the primary side of the power converter **700**, and a source

terminal of the transistor **1702** (e.g., JFET) is connected to one terminal of the resistor **2710** (e.g., a resistor R1), one terminal of the resistor **2720** (e.g., a resistor R2), and one terminal of the resistor **2730** (e.g., a resistor R3). For example, another terminal of the resistor **2710** (e.g., a resistor R1) is connected to one terminal of the switch **2712** (e.g., a switch S1), another terminal of the resistor **2720** (e.g., a resistor R2) is connected to one terminal of the switch **2722** (e.g., a switch S2), and another terminal of the resistor **2730** (e.g., a resistor R3) is connected to one terminal of the switch **2732** (e.g., a switch S3). As an example, another terminal of the switch **2712** (e.g., a switch S1), another terminal of the switch **2722** (e.g., a switch S2), and another terminal of the switch **2732** (e.g., a switch S3) are all connected to a source terminal of the transistor **1706** (e.g., a transistor M3). For example, the transistor **1706** (e.g., a transistor M3) is a PMOS transistor (e.g., a high-voltage PMOS transistor).

(116) In some examples, the switch **2712** (e.g., a switch S1) receives a signal **2714** that is used to close (e.g., turn on) or open (e.g., turn off) the switch **2712** (e.g., a switch S1), the switch **2722** (e.g., a switch S2) receives a signal **2724** that is used to close (e.g., turn on) or open (e.g., turn off) the switch **2722** (e.g., a switch S1), and the switch **2732** (e.g., a switch S3) receives a signal **2734** that is used to close (e.g., turn on) or open (e.g., turn off) the switch **2732** (e.g., a switch S1). In certain examples, a gate terminal of the transistor **1706** (e.g., a transistor M3) receives a signal **1732** from the charging controller **1730**. For example, a drain terminal of the transistor **1706** (e.g., a transistor M3) is connected to the terminal **748** (e.g., a VDD pin) and is also connected to one terminal of the inductor **716** through the terminal **748** (e.g., a VDD pin).

(117) In some embodiments, another terminal of the inductor **716** is connected to the terminal **732** (e.g., a SW pin) and is also connected to an anode of the diode **1782** (e.g., a diode D1) through the terminal **732** (e.g., a SW pin). For example, a cathode of the diode **1782** (e.g., a diode D1) is connected to the terminal **734** (e.g., a V10 pin) and one terminal of the resistor **1754** (e.g., a resistor R4). As an example, another terminal of the resistor **1754** (e.g., a resistor R4) is connected to an input terminal of the DC-DC controller **1750** and one terminal of the resistor **1756** (e.g., a resistor R5). For example, another terminal of the resistor **1756** (e.g., a resistor R5) is connected to the terminal **738** (e.g., a GND pin), which is biased to the ground voltage on the primary side of the power converter **700**. In certain examples, an output terminal of the DC-DC controller **1750** is connected to a gate terminal of the transistor **1752** (e.g., a transistor M4). For example, a source terminal of the transistor **1752** (e.g., a transistor M4) is biased to the ground voltage on the primary side of the power converter **700**. As an example, a drain terminal of the transistor **1752** (e.g., a transistor M4) is connected to the anode of the diode **1782** and the terminal **732** (e.g., a SW pin). For example, the transistor **1752** (e.g., a transistor M4) is an NMOS transistor.

(118) According to some embodiments, an input terminal of the voltage detection circuit **1720** is connected to the terminal **748** (e.g., a VDD pin). For example, the output terminal of the voltage detection circuit **1720** sends a signal **1721** to the charging controller **1730**. In some examples, an input terminal of the voltage detection circuit **1770** is connected to the terminal **734** (e.g., a V10 pin). For example, the voltage detection circuit **1770** generates signals **1722**, **1771**, **1773**, **2714**, **2724**, and **2734**. As an example, the voltage detection circuit **1770** sends the signals **1722** and **1771** to the charging controller **1730** and sends the signal **1773** to the circuit **1780** for undervoltage-lockout (UVLO). In certain examples, an output terminal of the circuit **1780** for undervoltage-lockout (UVLO) is connected to an input terminal of the internal control circuit **1790**. As an example, the internal control circuit **1790** includes various control circuits, and the internal control circuit **1790** does not include the charging controller **1730**. For example, the internal control circuit **1790** includes one or more control circuits for one or more oscillators, one or more control circuits for one or more timers, one or more drivers, one or more logic control circuits, and/or one or more protection circuits.

(119) As shown in FIG. 7, after the EMI filter **722** receives the AC voltage **750**, the PWM controller **730** (e.g., an IC chip) undergoes a startup process according to certain embodiments.

According to some embodiments, during the startup process, if the voltage detection circuit **1770** determines that the voltage **735** has not reached a startup threshold (e.g., **V10_UVLO**), the voltage detection circuit **1770** generates the signal **1722**, so that the charging controller **1730** generates the signal **1732** to turn on the transistor **1706** (e.g., a transistor **M3**). In certain examples, if the voltage detection circuit **1770** determines that the voltage **735** is smaller than a first reference voltage (e.g., **Vx**), the voltage detection circuit **1770** generates the signal **2714** to close (e.g., turn on) the switch **2712**, generates the signal **2724** to open (e.g., turn off) the switch **2722**, and generates the signal **2734** to open (e.g., turn off) the switch **2732**. In some examples, if the voltage detection circuit **1770** determines that the voltage **735** is larger than the first reference voltage (e.g., **Vx**) but smaller than a second reference voltage (e.g., **Vy**), the voltage detection circuit **1770** generates the signal **2714** to close (e.g., turn on) the switch **2712**, generates the signal **2724** to close (e.g., turn on) the switch **2722**, and generates the signal **2734** to open (e.g., turn off) the switch **2732**. For example, the first reference voltage (e.g., **Vx**) is smaller than the second reference voltage (e.g., **Vy**), and the second reference voltage (e.g., **Vy**) is smaller than the startup threshold (e.g., **V10_UVLO**). In certain examples, if the voltage detection circuit **1720** determines that the voltage **735** is larger than both the first reference voltage (e.g., **Vx**) and the second reference voltage (e.g., **Vy**), the voltage detection circuit **1720** generates the signal **2714** to close (e.g., turn on) the switch **2712**, generates the signal **2724** to close (e.g., turn on) the switch **2722**, and generates the signal **2734** to close (e.g., turn on) the switch **2732**.

(120) In certain embodiments, during the startup process, if the voltage **735** has not reached the startup threshold (e.g., **V10_UVLO**), the charging circuit **1710** generates a charging current **1792** to charge the capacitor **718** (e.g., the capacitor **C5**) and raise the voltage **735** at the terminal **734** (e.g., a **V10** pin). For example, the charging current **1792** is determined as follows:

$$(121) \quad I_{hv_v10} = \begin{cases} \frac{V_{HVS} - V_{10}}{R_1} & V_{10} < V_x \\ \frac{V_{HVS} - V_{10}}{R_1 // R_2} & V_x < V_{10} < V_y \\ \frac{V_{HVS} - V_{10}}{R_1 // R_2 // R_3} & V_y < V_{10} < V_{10_UVLO} \end{cases} \quad (\text{Equation 7})$$

where I_{hv_v10} represents the charging current **1792**. Additionally, V_{HVS} represents the voltage **741**, and V_{10} represents the voltage **735**. Moreover, $R_{sub.1}$ represents the resistance of the resistor **2710**, $R_{sub.2}$ represents the resistance of the resistor **2720**, and $R_{sub.3}$ represents the resistance of the resistor **2730**. Also, $R_{sub.1}/R_{sub.2}$ represents the resistance of a combination of the resistors **2710** and **2720**, which are connected in parallel, and $R_{sub.1}/R_{sub.2}/R_{sub.3}$ represents the resistance of a combination of the resistors **2710**, **2720** and **2730**, which are connected in parallel. Additionally, $V_{sub.x}$ represents the first reference voltage, $V_{sub.y}$ represents the second reference voltage, and **V10_UVLO** represents the startup threshold.

(122) In some embodiments, if the voltage **735** is smaller than the first reference voltage (e.g., **Vx**), the charging current **1792** flows through the transistor **1702** (e.g., FET), the resistor **2710** (e.g., a resistor **R1**), and the transistor **1706** (e.g., a transistor **M3**). For example, if the voltage **735** is larger than the first reference voltage (e.g., **Vx**) but smaller than the second reference voltage (e.g., **Vy**), the charging current **1792** flows through the transistor **1702** (e.g., FET), the resistor **2710** (e.g., a resistor **R1**), the resistor **2720** (e.g., a resistor **R2**), and the transistor **1706** (e.g., a transistor **M3**). As an example, if the voltage **735** is larger than both the first reference voltage (e.g., **Vx**) and the second reference voltage (e.g., **Vy**) but smaller than the startup threshold (e.g., **V10_UVLO**), the charging current **1792** flows through the transistor **1702** (e.g., FET), the resistor **2710** (e.g., a resistor **R1**), the resistor **2720** (e.g., a resistor **R2**), the resistor **2730** (e.g., a resistor **R3**), and the transistor **1706** (e.g., a transistor **M5**).

(123) According to certain embodiments, before the voltage **735** reaches the startup threshold (e.g., **V10_UVLO**), the voltage detection circuit **1770** generates the signal **1773**, so that the circuit **1780** for undervoltage-lockout (UVLO) does not generate the signal **1791** that indicates the end of the

startup process and causes the internal control circuit **1790** to power on other components of the PWM controller **730** (e.g., an IC chip). For example, during the startup process, other components of the PWM controller **730** (e.g., an IC chip) remain powered off. As an example, during the startup process, the DC-DC converter does not operate.

(124) According to some embodiments, if the voltage detection circuit **1770** determines that the voltage **735** has risen above the startup threshold (e.g., **V10_UVLO**), the voltage detection circuit **1770** generates the signal **1722**, so that the charging controller **1730** generates the signal **1732** to turn off the transistor **1706** (e.g., a transistor **M3**) and the charging circuit **1710** does not charge the capacitor **718** (e.g., a capacitor **C5**). In some examples, after the voltage detection circuit **1770** determines that the voltage **735** has risen above the startup threshold (e.g., **V10_UVLO**), the startup process of the PWM controller **730** (e.g., an IC chip) ends and the normal operation for the power converter **700** starts. For example, at the beginning of the normal operation, the voltage detection circuit **1770** generates the signal **1773**, so that the circuit **1780** for undervoltage-lockout (UVLO) generates the signal **1791** that indicates the end of the startup process and causes the internal control circuit **1790** to power on other components of the PWM controller **730** (e.g., an IC chip). As an example, during the normal operation, other components of the PWM controller **730** (e.g., an IC chip) remain powered on. In certain examples, during the startup process, the voltage detection circuit **1770** determines whether the voltage **735** is smaller than the first reference voltage (e.g., **Vx**), whether the voltage **735** is larger than the first reference voltage (e.g., **Vx**) but smaller than the second reference voltage (e.g., **Vy**), and/or whether the voltage **735** is larger than both the first reference voltage (e.g., **Vx**) and the second reference voltage (e.g., **Vy**) but smaller than the startup threshold (e.g., **V10_UVLO**). In some examples, during the normal operation, the voltage detection circuit **1770** determines whether the voltage **735** is smaller than the first reference voltage (e.g., **Vx**), whether the voltage **735** is larger than the first reference voltage (e.g., **Vx**) but smaller than the second reference voltage (e.g., **Vy**), and/or whether the voltage **735** is larger than both the first reference voltage (e.g., **Vx**) and the second reference voltage (e.g., **Vy**), and also determines whether the voltage **735** rises to a predetermined threshold (e.g., **V10_HV_OFF**).

(125) In certain embodiments, during the normal operation, when the voltage **749** (e.g., **V.sub.DD**) is higher than a reference voltage (e.g., **V.sub.reg**), the DC-DC converter is idle, and the auxiliary winding **706** charges the capacitor **768** (e.g., a capacitor **C3**) to support the voltage **749** (e.g., **V.sub.DD**), which is used to charge the capacitor **718** (e.g., a capacitor **C5**) and support the voltage **735** so that the voltage **735** at the terminal **734** (e.g., a **V10** pin) is equal to the voltage **749** (e.g., **V.sub.DD**) minus the forward bias voltage of the diode **1582**. In some examples, during the normal operation, when the voltage **749** (e.g., **V.sub.DD**) is smaller than the reference voltage (e.g., **V.sub.reg**) but is not smaller than a lower predetermined threshold (e.g., **VDD_HV_ON**), the voltage **735** at the terminal **734** (e.g., a **V10** pin) is generated by the DC-DC converter using the voltage **749** (e.g., **V.sub.DD**), and the generated voltage **735** is larger than the voltage **749** (e.g., **V.sub.DD**) and is equal to the reference voltage (e.g., **V.sub.reg**).

(126) According to some embodiments, during the normal operation, if the voltage detection circuit **1720** determines that the voltage **749** drops to the lower predetermined threshold (e.g., **VDD_HV_ON**), the voltage detection circuit **1720** generates the signal **1721**, so that the charging controller **1730** generates the signal **1732** to turn on the transistor **1706** (e.g., a transistor **M3**). In some examples, if the transistor **1706** (e.g., a transistor **M3**) is turned on, the charging circuit **1710** generates the charging current **1792** to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735**. For example, during the normal operation, when the transistor **1706** (e.g., a transistor **M3**) is turned on, if the voltage detection circuit **1770** determines the voltage **735** is smaller than the first reference voltage (e.g., **Vx**), the switch **2512** is closed but the switches **2522** and **2532** are open. As an example, during the normal operation, when the transistor **1706** (e.g., a transistor **M3**) is turned on, if the voltage **735** is larger than the first reference voltage (e.g., **Vx**) but smaller than

the second reference voltage (e.g., V_y), the switches **2512** and **2522** are closed but the switch **2532** is open. For example, during the normal operation, when the transistor **1706** (e.g., a transistor **M3**) is turned on, if the voltage **735** is larger than both the first reference voltage (e.g., V_x) and the second reference voltage (e.g., V_y), the switches **2512**, **2522**, and **2532** are closed. In certain examples, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1710** generates the charging current **1792** to flow through the terminal **734** (e.g., a **V10** pin) and to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735**, in order to prevent insufficient power supply to the PWM controller **730** (e.g., an IC chip).

(127) In certain embodiments, during the normal operation, if the voltage detection circuit **1720** determines that the voltage **749** rises to a higher predetermined threshold (e.g., $V_{DD_HV_OFF}$), the voltage detection circuit **1720** generates the signal **1721**, so that the charging controller **1730** generates the signal **1732** to turn off the transistor **1706** (e.g., a transistor **M3**). For example, if the transistor **1706** (e.g., a transistor **M3**) is turned off, the charging circuit **1710** does not charge the capacitor **718** (e.g., a capacitor **C5**). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1710** does not charge the capacitor **718** (e.g., a capacitor **C5**) in order to reduce power consumption of the PWM controller **730** (e.g., an IC chip).

(128) In some embodiments, during the normal operation, the voltage detection circuit **1770** determines whether the voltage **735** rises to the predetermined threshold (e.g., $V_{10_HV_OFF}$) and generates the signal **1771**. For example, if the voltage **735** has not risen to the predetermined threshold (e.g., $V_{10_HV_OFF}$), the signal **1771** is at a first logic level (e.g., a logic low level). As an example, if the voltage **735** has risen to the predetermined threshold (e.g., $V_{10_HV_OFF}$), the signal **1771** is at a second logic level (e.g., a logic high level). In certain examples, if the signal **1771** indicates that the voltage **735** has risen to the predetermined threshold (e.g., $V_{10_HV_OFF}$), the charging controller **1730** generates the signal **1732** to turn off the transistor **1706** (e.g., a transistor **M3**). For example, if the transistor **1706** (e.g., a transistor **M3**) is turned off, the charging circuit **1710** does not charge the capacitor **718** (e.g., a capacitor **C5**). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1710** does not charge the capacitor **718** (e.g., a capacitor **C5**) in order to reduce power consumption of the PWM controller **730** (e.g., an IC chip).

(129) As shown in FIG. 17, during the normal operation, the charging controller **1730** receives the signals **1721** and **1771** and generates the signal **1732** to turn on or turn off the transistor **1706** according to some embodiments. For example, if the signal **1721** indicates that the voltage **749** drops to the lower predetermined threshold (e.g., $V_{DD_HV_ON}$), the charging controller **1730** generates the signal **1732** to turn on the transistor **1706** (e.g., a transistor **M3**) for generating the charging current **1792** to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735**. As an example, if the signal **1721** indicates that the voltage **749** rises to the higher predetermined threshold (e.g., $V_{DD_HV_OFF}$), the charging controller **1730** generates the signal **1732** to turn off the transistor **1706** (e.g., a transistor **M3**). For example, if the signal **1771** indicates that the voltage **735** has risen to the predetermined threshold (e.g., $V_{10_HV_OFF}$), the charging controller **1730** generates the signal **1732** to turn off the transistor **1706** (e.g., a transistor **M3**).

(130) As discussed above and further emphasized here, FIG. 17 is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. In some examples, the PWM controller **730** also includes a reference-current and reference-voltage generator that receives a signal from the voltage detection circuit **1770** and is connected to the internal control circuit **1790**. In certain examples, the voltage detection circuit **1770** outputs a signal for overvoltage protection. In certain examples, the diode **1782** (e.g., a Schottky diode) is replaced by a parasitic body diode of a PMOS transistor and/or by a PMOS switch. In some examples, the PWM controller **730** also includes a timer that, after being triggered, during a predetermined duration, prevents the signal **1771** from causing the

charging controller 1730 to turn off the transistor 1706 (e.g., a transistor M3).

(131) According to some embodiments, a controller for a power converter includes: a first terminal configured to receive a first voltage; a second terminal connected to a capacitor and biased to a second voltage; a voltage detector configured to receive the second voltage from the second terminal and generate a detection signal based at least in part on the second voltage; a charging controller configured to receive the detection signal and generate a first control signal based at least in part on the detection signal; and a charging current generator configured to receive the first voltage from the first terminal and receive the first control signal from the charging controller; wherein the voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the charging controller is further configured to: in response to the detection signal indicating that the second voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a charging current is to be generated; wherein the charging current generator is further configured to: in response to the first control signal indicating that the charging current is to be generated, generate the charging current that flows through the second terminal to charge the capacitor associated with the second voltage. For example, the controller is implemented according to at least FIG. 9.

(132) As an example, the voltage detector is further configured to: detect that the second voltage has increased to a second predetermined threshold; and generate the detection signal indicating that the second voltage has increased to the second predetermined threshold; wherein the second predetermined threshold is larger than the first predetermined threshold. For example, the charging controller is further configured to: in response to the detection signal indicating that the second voltage has increased to the second predetermined threshold, generate the first control signal indicating that the charging current is not to be generated; wherein the charging current generator is further configured to: in response to the first control signal indicating that the charging current is not to be generated, not generate the charging current that flows through the second terminal.

(133) As an example, the second voltage decreases to the first predetermined threshold during normal operation after a startup process; and the second voltage increases to the second predetermined threshold during the normal operation. For example, during the startup process: the voltage detector is further configured to: detect that the second voltage has not risen above a third predetermined threshold; and generate the detection signal indicating that the second voltage has not risen above the third predetermined threshold; the charging controller is further configured to: in response to the detection signal indicating that the second voltage has not risen above the third predetermined threshold, generate the first control signal indicating that the charging current is to be generated; wherein the charging current generator is further configured to: in response to the first control signal indicating that the charging current is to be generated, generate the charging current that flows through the second terminal to charge the capacitor and raise the second voltage. As an example, during the startup process: the voltage detector is further configured to: detect that the second voltage has risen above the third predetermined threshold; and generate the detection signal indicating that the second voltage has risen above the third predetermined threshold; the charging controller is further configured to: in response to the detection signal indicating that the second voltage has risen above the third predetermined threshold, generate the first control signal indicating that the charging current is not to be generated; wherein the charging current generator is further configured to: in response to the first control signal indicating that the charging current is not to be generated, not generate the charging current that flows through the second terminal to charge the capacitor and raise the second voltage. For examples, the controller further includes: a circuit for undervoltage-lockout configured to: in response to the second voltage has risen above the third predetermined threshold, generating a second control signal indicating the startup process has ended; and an internal control circuit configured to: in response to the second control signal

indicating the startup process has ended, power on one or more components of the controller to start the normal operation.

(134) According to certain embodiments, a controller for a power converter includes: a first terminal configured to receive a first voltage; a second terminal connected to a first capacitor and biased to a second voltage; a third terminal connected to a second capacitor and biased to a third voltage; a first voltage detector configured to receive the second voltage from the second terminal and generate a first detection signal based at least in part on the second voltage; a first charging controller configured to receive the first detection signal and generate a first control signal based at least in part on the first detection signal; and a charging current generator configured to receive the first voltage from the first terminal, the charging current generator including a first current generator and a second current generator, the first current generator being configured to receive the first control signal from the first charging controller; wherein the first voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the first charging controller is further configured to: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a first charging current is to be generated; wherein the first current generator is further configured to: in response to the first control signal indicating that the first charging current is to be generated, generate the first charging current that flows through the third terminal to charge the second capacitor associated with the third voltage. For example, the controller is implemented according to at least FIG. 13 and/or FIG. 15.

(135) As an example, the first voltage detector is further configured to: detect that the second voltage has increased to a second predetermined threshold; and generate the first detection signal indicating that the second voltage has increased to the second predetermined threshold; wherein the second predetermined threshold is larger than the first predetermined threshold. For example, the controller further includes: a second voltage detector configured to receive the third voltage from the third terminal and generate a second detection signal based at least in part on the third voltage; wherein the second voltage detector is further configured to: detect that the third voltage has increased to a third predetermined threshold; and generate the second detection signal indicating that the third voltage has increased to the third predetermined threshold. As an example, the first charging controller is further configured to: in response to the first detection signal indicating that the second voltage has increased to the second predetermined threshold, generate the first control signal indicating that the first charging current is not to be generated; and in response to the second detection signal indicating that the third voltage has increased to the third predetermined threshold, generate the first control signal indicating that the first charging current is not to be generated; wherein the first current generator is further configured to: in response to the first control signal indicating that the first charging current is not to be generated, not generate the first charging current that flows through the third terminal.

(136) For example, the second voltage decreases to the first predetermined threshold during normal operation after a startup process; the second voltage increases to the second predetermined threshold during the normal operation; and the third voltage increases to the third predetermined threshold during the normal operation. As an example, during the startup process: the second voltage detector is further configured to: detect that the third voltage has not risen above a fourth predetermined threshold; and generate a third detection signal indicating that the third voltage has not risen above the fourth predetermined threshold. For example, the controller further includes a second charging controller configured to receive the third detection signal and generate a second control signal based at least in part on the third detection signal. As an example, the second charging controller is further configured to: in response to the third detection signal indicating that the third voltage has not risen above the fourth predetermined threshold, generate the second

control signal indicating that a second charging current is to be generated; wherein the second current generator is further configured to: receive the second control signal from the second charging controller; and in response to the second control signal indicating that the second charging current is to be generated, generate the second charging current that flows through the second terminal.

(137) For example, the first voltage detector is further configured to: determine whether the second voltage is smaller than a first reference voltage, whether the second voltage is larger than the first reference voltage but smaller than a second reference voltage, and whether the second voltage is larger than the second reference voltage; generate a fourth detection signal indicating that the second voltage is smaller than the first reference voltage; generate a fifth detection signal indicating that the second voltage is larger than the first reference voltage but smaller than the second reference voltage; and generate a sixth detection signal indicating that the second voltage is larger than the second reference voltage. As an example, the second current generator is further configured to, in response to the second control signal indicating that the second charging current is to be generated: generate the second charging current that flows through a first resistor but not a second resistor and not a third resistor, if the second current generator receives the fourth detection signal indicating that the second voltage is smaller than the first reference voltage; generate the second charging current that flows through the first resistor and the second resistor but not the third resistor, if the second current generator receives the fifth detection signal indicating that the second voltage is larger than the first reference voltage but smaller than the second reference voltage; and generate the second charging current that flows through the first resistor, the second resistor and the third resistor, if the second current generator receives the sixth detection signal indicating that the second voltage is larger than the second reference voltage. For example, the controller further includes: a circuit for undervoltage-lockout configured to: in response to the third voltage has risen above the fourth predetermined threshold, generating a third control signal indicating the startup process has ended; and an internal control circuit configured to: in response to the third control signal indicating the startup process has ended, power on one or more components of the controller to start the normal operation.

(138) According to some embodiments, a controller for a power converter includes: a first terminal configured to receive a first voltage; a second terminal connected to a first capacitor and biased to a second voltage; a third terminal connected to a second capacitor and biased to a third voltage; a first voltage detector configured to receive the second voltage from the second terminal and generate a first detection signal based at least in part on the second voltage; a charging controller configured to receive the first detection signal and generate a first control signal based at least in part on the first detection signal; and a charging current generator configured to receive the first voltage from the first terminal and configured to receive the first control signal from the charging controller; wherein the first voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the charging controller is further configured to: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a charging current is to be generated; wherein the current generator is further configured to: in response to the first control signal indicating that the charging current is to be generated, generate the charging current that flows through the third terminal to charge the second capacitor associated with the third voltage. For example, the controller is implemented according to at least FIG. 17.

(139) As an example, the first voltage detector is further configured to: detect that the second voltage has increased to a second predetermined threshold; and generate the first detection signal indicating that the second voltage has increased to the second predetermined threshold; wherein the second predetermined threshold is larger than the first predetermined threshold. For example, the

controller further includes: a second voltage detector configured to receive the third voltage from the third terminal and generate a second detection signal based at least in part on the third voltage; wherein the second voltage detector is further configured to: detect that the third voltage has increased to a third predetermined threshold; and generate the second detection signal indicating that the third voltage has increased to the third predetermined threshold. As an example, the charging controller is further configured to: in response to the first detection signal indicating that the second voltage has increased to the second predetermined threshold, generate the first control signal indicating that the charging current is not to be generated; and in response to the second detection signal indicating that the third voltage has increased to the third predetermined threshold, generate the first control signal indicating that the charging current is not to be generated; wherein the charging current generator is further configured to: in response to the first control signal indicating that the charging current is not to be generated, not generate the charging current that flows through the third terminal.

(140) For example, the second voltage decreases to the first predetermined threshold during normal operation after a startup process; the second voltage increases to the second predetermined threshold during the normal operation; and the third voltage increases to the third predetermined threshold during the normal operation. As an example, during the startup process: the second voltage detector is further configured to: detect that the third voltage has not risen above a fourth predetermined threshold; and generate a third detection signal indicating that the third voltage has not risen above the fourth predetermined threshold. For example, the charging controller is further configured to: in response to the third detection signal indicating that the third voltage has not risen above the fourth predetermined threshold, generate the first control signal indicating that the charging current is to be generated; wherein the current generator is further configured to: in response to the first control signal indicating that the charging current is to be generated, generate the charging current that flows through the third terminal to charge the second capacitor associated with the third voltage.

(141) As an example, the second voltage detector is further configured to: determine whether the third voltage is smaller than a first reference voltage, whether the third voltage is larger than the first reference voltage but smaller than a second reference voltage, and whether the third voltage is larger than the second reference voltage; generate a fourth detection signal indicating that the third voltage is smaller than the first reference voltage; generate a fifth detection signal indicating that the third voltage is larger than the first reference voltage but smaller than the second reference voltage; and generate a sixth detection signal indicating that the third voltage is larger than the second reference voltage. For example, wherein the current generator is further configured to, in response to the first control signal indicating that the charging current is to be generated: generate the charging current that flows through a first resistor but not a second resistor and not a third resistor, if the current generator receives the fourth detection signal indicating that the third voltage is smaller than the first reference voltage; generate the charging current that flows through the first resistor and the second resistor but not the third resistor, if the current generator receives the fifth detection signal indicating that the third voltage is larger than the first reference voltage but smaller than the second reference voltage; and generate the charging current that flows through the first resistor, the second resistor and the third resistor, if the current generator receives the sixth detection signal indicating that the third voltage is larger than the second reference voltage. As an example, the controller further includes: a circuit for undervoltage-lockout configured to: in response to the third voltage has risen above the fourth predetermined threshold, generating a second control signal indicating the startup process has ended; and an internal control circuit configured to: in response to the second control signal indicating the startup process has ended, power on one or more components of the controller to start the normal operation.

(142) According to certain embodiments, a method for a power converter includes: receiving a first voltage; receiving a second voltage associated with a capacitor; generating a detection signal based

at least in part on the second voltage; receiving the detection signal; generating a first control signal based at least in part on the detection signal; receiving the first control signal; and processing the first control signal; wherein the generating a detection signal based at least in part on the second voltage includes: detecting that the second voltage has decreased to a first predetermined threshold; and generating the detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the generating a first control signal based at least in part on the detection signal includes: in response to the detection signal indicating that the second voltage has decreased to the first predetermined threshold, generating the first control signal indicating that a charging current is to be generated; wherein the processing the first control signal includes: in response to the first control signal indicating that the charging current is to be generated, generating the charging current to charge the capacitor associated with the second voltage. For example, the method is implemented according to at least FIG. 9.

(143) According to some embodiments, a method for a power converter includes: receiving a first voltage; receiving a second voltage associated with a first capacitor; receiving a third voltage associated with a second capacitor; generating a first detection signal based at least in part on the second voltage; receiving the first detection signal; generating a first control signal based at least in part on the first detection signal; receiving the first control signal; and processing the first control signal; wherein the generating a first detection signal based at least in part on the second voltage includes: detecting that the second voltage has decreased to a first predetermined threshold; and generating the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the generating a first control signal based at least in part on the first detection signal includes: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generating the first control signal indicating that a first charging current is to be generated; wherein the processing the first control signal includes: in response to the first control signal indicating that the first charging current is to be generated, generating the first charging current to charge the second capacitor associated with the third voltage. For example, the method is implemented according to at least FIG. 13 and/or FIG. 15.

(144) According to certain embodiments, a method for a power converter includes: receiving a first voltage; receiving a second voltage associated with a first capacitor; receiving a third voltage associated with a second capacitor; generating a first detection signal based at least in part on the second voltage; receiving the first detection signal; generating a first control signal based at least in part on the first detection signal; receiving the first control signal; and processing the first control signal; wherein the generating a first detection signal based at least in part on the second voltage includes: detecting that the second voltage has decreased to a first predetermined threshold; and generating the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the generating a first control signal based at least in part on the first detection signal includes: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generating the first control signal indicating that a charging current is to be generated; wherein the processing the first control signal includes: in response to the first control signal indicating that the charging current is to be generated, generating the charging current to charge the second capacitor associated with the third voltage. For example, the method is implemented according to at least FIG. 17.

(145) For example, some or all components of various embodiments of the present invention each are, individually and/or in combination with at least another component, implemented using one or more software components, one or more hardware components, and/or one or more combinations of software and hardware components. As an example, some or all components of various embodiments of the present invention each are, individually and/or in combination with at least another component, implemented in one or more circuits, such as one or more analog circuits and/or one or more digital circuits. For example, various embodiments and/or examples of the

present invention can be combined.

(146) Although specific embodiments of the present invention have been described, it will be understood by those of skill in the art that there are other embodiments that are equivalent to the described embodiments. Accordingly, it is to be understood that the invention is not to be limited by the specific illustrated embodiments.

Claims

1. A converter controller for a power converter, the converter controller comprising: a voltage detector configured to receive a voltage and generate a detection signal based at least in part on the voltage; and a charging controller configured to receive the detection signal and generate a first control signal based at least in part on the detection signal; wherein the voltage detector is further configured to: detect that the voltage has decreased to a first predetermined threshold; and generate the detection signal indicating that the voltage has decreased to the first predetermined threshold; wherein the charging controller is further configured to: in response to the detection signal indicating that the voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a charging current is to be generated for raising the voltage; wherein the voltage detector is further configured to: detect that the voltage has increased to a second predetermined threshold; and generate the detection signal indicating that the voltage has increased to the second predetermined threshold; wherein the second predetermined threshold is larger than the first predetermined threshold.
2. The converter controller of claim 1 wherein: the charging controller is further configured to: in response to the detection signal indicating that the voltage has increased to the second predetermined threshold, generate the first control signal indicating that the charging current is not to be generated.
3. The converter controller of claim 1 wherein: the voltage decreases to the first predetermined threshold during normal operation after a startup process; and the voltage increases to the second predetermined threshold during the normal operation.
4. The converter controller of claim 3 wherein, during the startup process: the voltage detector is further configured to: detect that the voltage has not reached a third predetermined threshold; and generate the detection signal indicating that the voltage has not reached the third predetermined threshold; the charging controller is further configured to: in response to the detection signal indicating that the voltage has not risen above the third predetermined threshold, generate the first control signal indicating that the charging current is to be generated for raising the voltage.
5. The converter controller of claim 4 wherein, during the startup process: the voltage detector is further configured to: detect that the voltage has risen above the third predetermined threshold; and generate the detection signal indicating that the voltage has risen above the third predetermined threshold; the charging controller is further configured to: in response to the detection signal indicating that the voltage has risen above the third predetermined threshold, generate the first control signal indicating that the charging current is not to be generated.
6. The converter controller of claim 5, and further comprising: a circuit for undervoltage-lockout configured to: in response to the voltage has risen above the third predetermined threshold, generating a second control signal indicating the startup process has ended; and an internal control circuit configured to: in response to the second control signal indicating the startup process has ended, power on one or more components of the converter controller to start the normal operation.
7. A converter controller for a power converter, the converter controller comprising: a first voltage detector configured to receive a first voltage and generate a first detection signal based at least in part on the first voltage; a second voltage detector configured to receive a second voltage and generate a second detection signal based at least in part on the second voltage; and a first charging controller configured to receive the first detection signal and the second detection signal and

generate a first control signal based at least in part on the first detection signal and the second detection signal; wherein the first voltage detector is further configured to: detect that the first voltage has decreased to a first predetermined threshold; and generate the first detection signal indicating that the first voltage has decreased to the first predetermined threshold; wherein the first charging controller is further configured to: in response to the first detection signal indicating that the first voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a first charging current is to be generated; wherein the second voltage detector is further configured to: detect that the second voltage has increased to a second predetermined threshold; and generate the second detection signal indicating that the second voltage has increased to the second predetermined threshold; wherein the first voltage and the second voltage are two different voltages.

8. The converter controller of claim 7 wherein the first charging controller is further configured to: in response to the second detection signal indicating that the second voltage has increased to the second predetermined threshold, generate the first control signal indicating that the first charging current is not to be generated.

9. The converter controller of claim 8, and further comprising: a first current generator configured to receive the first control signal from the first charging controller; wherein the first current generator is further configured to: in response to the first control signal indicating that the first charging current is to be generated, generate the first charging current for raising the second voltage; and in response to the first control signal indicating that the first charging current is not to be generated, not generate the first charging current.

10. The converter controller of claim 7 wherein the first voltage detector is further configured to: detect that the first voltage has increased to a third predetermined threshold; and generate the first detection signal indicating that the first voltage has increased to the third predetermined threshold; wherein the third predetermined threshold is larger than the first predetermined threshold.

11. The converter controller of claim 10 wherein the first charging controller is further configured to: in response to the first detection signal indicating that the first voltage has increased to the third predetermined threshold, generate the first control signal indicating that the first charging current is not to be generated.

12. The converter controller of claim 10 wherein the first voltage decreases to the first predetermined threshold during normal operation after a startup process.

13. The converter controller of claim 12 wherein the second voltage increases to the second predetermined threshold during the normal operation.

14. The converter controller of claim 12 wherein the first voltage increases to the third predetermined threshold during the normal operation.

15. A converter controller for a power converter, the converter controller comprising: a first voltage detector configured to receive a first voltage and generate a first detection signal based at least in part on the first voltage; a second voltage detector configured to receive a second voltage and generate a second detection signal based at least in part on the second voltage; a first charging controller configured to receive the first detection signal and generate a first control signal based at least in part on the first detection signal; and a second charging controller configured to receive the second detection signal and generate a second control signal based at least in part on the second detection signal; wherein, during a startup process before normal operation, the second voltage detector is further configured to: detect that the second voltage has not reached a first predetermined threshold; and generate the second detection signal indicating that the second voltage has not reached the first predetermined threshold; the second charging controller is further configured to: in response to the second detection signal indicating that the second voltage has not reached the first predetermined threshold, generate the second control signal indicating that a first charging current is to be generated to raise the first voltage; wherein, during the normal operation after the startup process, the first voltage detector is further configured to: detect that the first

voltage has decreased to a second predetermined threshold; and generate the first detection signal indicating that the first voltage has decreased to the second predetermined threshold; the first charging controller is further configured to: in response to the first detection signal indicating that the first voltage has decreased to the second predetermined threshold, generate the first control signal indicating that a second charging current is to be generated to raise the second voltage.

16. The converter controller of claim 15, and further comprising: a first current generator, the first current generator being configured to: receive the first control signal from the first charging controller; and in response to the first control signal indicating that the second charging current is to be generated, generate the second charging current to raise the second voltage.

17. The converter controller of claim 16, and further comprising: a second current generator, the second current generator being configured to: receive the second control signal from the second charging controller; and in response to the second control signal indicating that the first charging current is to be generated, generate the first charging current to raise the first voltage.

18. The converter controller of claim 17 wherein, during the startup process: the second voltage detector is further configured to: detect that the second voltage has risen above the first predetermined threshold; and generate the second detection signal indicating that the second voltage has risen above the first predetermined threshold.

19. The converter controller of claim 18 wherein the second charging controller is further configured to: in response to the second detection signal indicating that the second voltage has risen above the first predetermined threshold, generate the second control signal indicating that the first charging current is not to be generated.

20. The converter controller of claim 19 wherein the second current generator is further configured to, in response to the second control signal indicating that the first charging current is not to be generated, not generate the first charging current.

21. The converter controller of claim 18, and further comprising: a circuit for undervoltage-lockout configured to: in response to the second voltage has risen above the first predetermined threshold, generating a third control signal indicating that the startup process has ended; and an internal control circuit configured to: in response to the third control signal indicating that the startup process has ended, power on one or more components of the converter controller to start the normal operation.

22. A converter controller for a power converter, the converter controller comprising: a first voltage detector configured to receive a first voltage and generate a first detection signal based at least in part on the first voltage; a second voltage detector configured to receive a second voltage and generate a second detection signal based at least in part on the second voltage; a charging controller configured to receive the first detection signal and the second detection signal and generate a control signal based at least in part on the second detection signal; and a current generator configured to receive the control signal from the charging controller; wherein, during a startup process before normal operation, the second voltage detector is further configured to: detect that the second voltage has not reached a predetermined threshold; and generate the second detection signal indicating that the second voltage has not reached the predetermined threshold; the charging controller is further configured to: in response to the second detection signal indicating that the second voltage has not reached the predetermined threshold, generate the control signal indicating that a charging current is to be generated to raise the first voltage; the current generator is further configured to: in response to the control signal indicating that the charging current is to be generated, generate the charging current to raise the first voltage; wherein, during the startup process before the normal operation, the first voltage detector is further configured to: determine that the first voltage is smaller than a first reference voltage, that the first voltage is larger than the first reference voltage but smaller than a second reference voltage, or that the first voltage is larger than the second reference voltage; generate a third detection signal indicating that the first voltage is smaller than the first reference voltage, that the first voltage is larger than the first reference

voltage but smaller than the second reference voltage, or that the first voltage is larger than the second reference voltage; wherein the second reference voltage is larger than the first reference voltage.

23. The converter controller of claim 22 wherein the current generator is further configured to receive the third detection signal indicating that the first voltage is smaller than the first reference voltage, that the first voltage is larger than the first reference voltage but smaller than the second reference voltage, or that the first voltage is larger than the second reference voltage.

24. The converter controller of claim 23 wherein the current generator is further configured to: in response to the third detection signal indicating that the first voltage is smaller than the first reference voltage, generate the charging current that flows through a first resistor but not a second resistor and a third resistor; in response to the third detection signal indicating that the first voltage is larger than the first reference voltage but smaller than the second reference voltage, generate the charging current that flows through the first resistor and the second resistor but not the third resistor; and in response to the third detection signal indicating that the first voltage is larger than the second reference voltage, generate the charging current that flows through the first resistor, the second resistor and the third resistor.

25. A converter controller for a power converter, the converter controller comprising: a first voltage detector configured to receive a first voltage and generate a first detection signal based at least in part on the first voltage; a second voltage detector configured to receive a second voltage and generate a second detection signal based at least in part on the second voltage; and a charging controller configured to receive the first detection signal and the second detection signal and generate a control signal based at least in part on the first detection signal and the second detection signal; wherein, during a startup process before normal operation, the second voltage detector is further configured to: detect that the second voltage has not reached a first predetermined threshold; and generate the second detection signal indicating that the second voltage has not reached the first predetermined threshold; the charging controller is further configured to: in response to the second detection signal indicating that the second voltage has not reached the first predetermined threshold, generate the control signal indicating that a charging current is to be generated to raise the second voltage; wherein, during the normal operation after the startup process, the first voltage detector is further configured to: detect that the first voltage has decreased to a second predetermined threshold; and generate the first detection signal indicating that the first voltage has decreased to the second predetermined threshold; the charging controller is further configured to: in response to the first detection signal indicating that the first voltage has decreased to the second predetermined threshold, generate the control signal indicating that the charging current is to be generated to raise the second voltage.

26. A converter controller for a power converter, the converter controller comprising: a first voltage detector configured to receive a first voltage and generate a first detection signal based at least in part on the first voltage; a second voltage detector configured to receive a second voltage and generate a second detection signal based at least in part on the second voltage; a charging controller configured to receive the first detection signal and the second detection signal and generate a control signal based at least in part on the second detection signal; and a current generator configured to receive the control signal from the charging controller; wherein, during a startup process before normal operation, the second voltage detector is further configured to: detect that the second voltage has not reached a predetermined threshold; and generate the second detection signal indicating that the second voltage has not reached the predetermined threshold; the charging controller is further configured to: in response to the second detection signal indicating that the second voltage has not reached the predetermined threshold, generate the control signal indicating that a charging current is to be generated to raise the second voltage; the current generator is further configured to: in response to the control signal indicating that the charging current is to be generated, generate the charging current to raise the second voltage; wherein, during the startup

process before the normal operation, the second voltage detector is further configured to: determine that the second voltage is smaller than a first reference voltage, that the second voltage is larger than the first reference voltage but smaller than a second reference voltage, or that the second voltage is larger than the second reference voltage but smaller than the predetermined threshold; and generate a third detection signal indicating that the second voltage is smaller than the first reference voltage, that the second voltage is larger than the first reference voltage but smaller than the second reference voltage, or that the second voltage is larger than the second reference voltage but smaller than the predetermined threshold; wherein: the second reference voltage is larger than the first reference voltage; and the predetermined threshold is larger than the second reference voltage.

27. The converter controller of claim 26 wherein the current generator is further configured to receive the third detection signal indicating that the second voltage is smaller than the first reference voltage, that the second voltage is larger than the first reference voltage but smaller than the second reference voltage, or that the second voltage is larger than the second reference voltage but smaller than the predetermined threshold.

28. The converter controller of claim 27 wherein the current generator is further configured to: in response to the third detection signal indicating that the second voltage is smaller than the first reference voltage, generate the charging current that flows through a first resistor but not a second resistor and a third resistor; in response to the third detection signal indicating that the second voltage is larger than the first reference voltage but smaller than the second reference voltage, generate the charging current that flows through the first resistor and the second resistor but not the third resistor; and in response to the third detection signal indicating that the second voltage is larger than the second reference voltage but smaller than the predetermined threshold, generate the charging current that flows through the first resistor, the second resistor and the third resistor.
